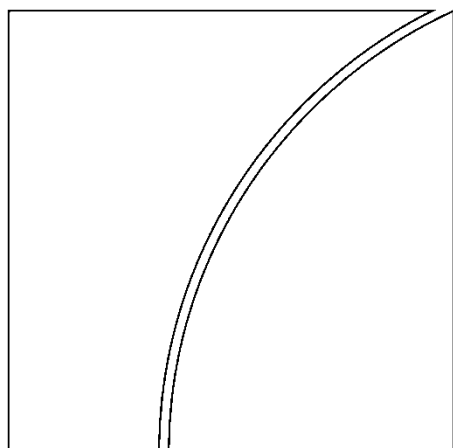




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by Lukas Menkhoff, Lucio Sarno, Maik Schmeling and Andreas Schrimpf

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Currency Momentum Strategies*

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Abstract

We provide a broad empirical investigation of momentum strategies in the foreign exchange market. We find a significant cross-sectional spread in excess returns of up to 10% p.a. between past winner and loser currencies. This spread in excess returns is not explained by traditional risk factors, it is partially explained by transaction costs and shows behavior consistent with investor under- and over-reaction. Moreover, cross-sectional currency momentum has very different properties from the widely studied carry trade and is not highly correlated with returns of benchmark technical trading rules. However, there seem to be very effective limits to arbitrage which prevent momentum returns from being easily exploitable in currency markets.

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1. Introduction

Momentum returns in stock markets provide a strong challenge to standard finance theory. Simply buying assets with high recent returns and selling assets with low recent returns results in a very profitable investment strategy whose returns are difficult to understand by means of standard risk factors (Jegadeesh and Titman, 1993, 2001). Consequently, researchers have proposed various explanations which focus not only on conventional risk-based models (e.g. Harvey and Siddique, 2000; Chordia and Shivakumar, 2002; Johnson, 2002; Pastor and Stambaugh, 2003; Liu and Zhang, 2011), but also on characteristics such as credit risk (Avramov, Chordia, Jostova, and Philipov, 2007) or bankruptcy risk (Eisdorfer, 2008), limits to arbitrage (e.g. Chabot, Ghysels, and Jagannathan, 2009), behavioral explanations such as investor under-reaction (e.g. Chui, Titman, and Wei, 2010), or high transaction costs (Korajczyk and Sadka, 2004). Despite this progress, the literature does not seem to have settled on a generally accepted explanation for momentum returns yet.

In this paper, we study foreign exchange (FX) markets as a natural laboratory for the analysis of momentum returns. Compared to stock markets, FX markets are more liquid and feature huge transaction volumes and low transaction costs, they are populated largely by sophisticated professional investors, and there are no natural short-selling constraints that prevent the shorting of past loser assets to fully implement momentum strategies. Hence, considering FX markets raises the hurdle for generating significant excess returns from momentum strategies considerably.

Surprisingly, there is little evidence on momentum in the *cross-section* of currencies. Large cross-country data sets were rare in the past so that the earlier literature has generally focused on momentum strategies in the time-series of currencies, i.e. momentum strategies where individual currencies are bought and sold over time depending on various sorts of signals such as moving average cross-overs, filter rules, channel breakouts, etc. This literature has shown that certain technical trading rules were temporarily profitable but that their profits often tend to deteriorate over time as more traders learn about these strategies and start to

exploit them (e.g., [Levich and Thomas III, 1993](#); [Pukthuanthong-Le, Levich, and Thomas III, 2007](#); [Neely, Weller, and Ulrich, 2009](#), among others). A survey of this literature is provided by [Menkhoff and Taylor \(2007\)](#). However, some evidence on the existence of cross-sectional momentum profits in the FX market is provided by [Okunev and White \(2003\)](#), [Asness, Moskowitz, and Pedersen \(2009\)](#) and [Burnside, Eichenbaum, and Rebelo \(2011\)](#) in the context of small cross sections of major currencies. Relative to our paper, these studies have a different focus, however, and do not provide a unifying analysis for understanding returns to cross-sectional currency momentum returns.

The main contribution of this paper is to study the economic anatomy of momentum profits in FX markets. We start by forming currency portfolios where an investor is long in currencies with high past excess returns (so-called “winners”) and short in currencies with low past excess returns (so-called “losers”). We take the viewpoint of a U.S. investor and consider exchange rates against the U.S. dollar (USD). Our data cover the period from January 1976 to January 2010, and we study a cross-section of up to 48 currencies. We go beyond earlier research on currency momentum by (a) providing an in-depth analysis of the relative importance of systematic versus unsystematic risk for understanding momentum returns, (b) carefully comparing momentum strategies to carry trades and technical trading rules, (c) quantifying the importance of transaction costs, and investigating non-standard sources of momentum returns, such as (d) under- and over-reaction or (e) limits to arbitrage.

We find large and significant excess returns to currency momentum strategies of up to 10% per annum (p.a.). As in [Jegadeesh and Titman \(2001\)](#), we find some evidence of return continuation and subsequent reversals over longer horizons of up to 36 months, which is consistent with behavioral biases, such as investor under- and over-reaction, and suggests that momentum effects in different asset classes may share a common source. Importantly, currency momentum is very different from the popular carry trade in FX markets, providing

high returns which are largely unrelated to carry trade returns.¹ Currency momentum returns are also different from returns generated by technical trading rules, which have been studied in a large empirical literature (e.g. (e.g. [Dooley and Shafer, 1976](#); [Sweeney, 1986](#); [Levich and Thomas III, 1993](#); [Neely, Weller, and Ulrich, 2009](#)).

In order to rationalize these high excess returns of currency momentum strategies, we investigate whether currency momentum is significantly affected by (i) transaction costs, (ii) business cycle risk and other traditional risk factors, and (iii) different forms of limits to arbitrage. We find that momentum returns are indeed fairly sensitive to transaction costs. Adjusting returns for bid-ask spreads lowers the profitability of momentum strategies significantly since momentum portfolios are skewed towards currencies with high transaction costs. However, transaction costs are unable to completely account for currency momentum returns.

Also, momentum returns in FX markets are not systematically related to standard proxies for business cycle risk, liquidity risk ([Brunnermeier, Nagel, and Pedersen, 2009](#)), the carry trade risk factor proposed by [Lustig, Roussanov, and Verdelhan \(2011\)](#), volatility risk ([Menkhoff, Sarno, Schmeling, and Schrimpf, 2011](#)), the three Fama-French factors ([Fama and French, 1992](#)) or a four-factor model including a U.S. stock return momentum factor ([Carhart, 1997](#)). In short, there does not seem to be a systematic risk factor which would explain (net) momentum returns, a result which is akin to the corresponding findings based on U.S. equity momentum.

However, the profitability of currency momentum strategies varies significantly over time, which may induce limits to arbitrage for the major market participants in FX markets (e.g. proprietary traders and hedge funds), who usually have rather short investment horizons

¹The carry trade is a popular trading strategy that borrows in currencies with low interest rates and invests in currencies with high interest rates. According to uncovered interest parity, if investors are risk neutral and form expectations rationally, exchange rate changes will eliminate any gain arising from the differential in interest rates across countries. However, a number of empirical studies show that high interest rate currencies tend to appreciate, while low interest rate currencies tend to depreciate. As a consequence, carry trades form a profitable investment strategy, giving rise to the “forward premium puzzle” ([Fama, 1984](#)). See [Burnside, Eichenbaum, Kleshchelski, and Rebelo \(2011\)](#), [Lustig, Roussanov, and Verdelhan \(2011\)](#), and [Menkhoff, Sarno, Schmeling, and Schrimpf \(2011\)](#).

and may thus act myopically (e.g. [Shleifer and Vishny, 1997](#)).² Furthermore, momentum returns are clearly related to currency characteristics. Returns are much higher in currencies with high (lagged) idiosyncratic volatility (about 8% p.a.) compared to currencies with low idiosyncratic volatility (about 4% p.a.). Returns are also related to measures of country risk, i.e. momentum strategies in countries with a high risk rating tend to yield significantly positive excess returns, whereas momentum strategies in countries with low risk ratings do not. Finally, a similar effect is found for a measure of exchange rate stability risk (i.e. the expected risk of observing large currency movements in the future).

In summary, we provide evidence that, despite FX markets' differences relative to stock markets, the properties of momentum strategies are fairly similar, which suggests that momentum profits in different asset classes may share a common root. Similar to stock markets, the high excess returns of currency momentum strategies can be (only) partially explained by their sensitivity to high transaction costs. Another piece of explanation of why momentum in currency markets persists is that there might be effective obstacles constraining the deployment of arbitrage capital to exploit the phenomenon. We find that currency momentum strategies are risky in that their returns are rather unstable over short time periods and that their exposure is subject to fundamental investment risk, captured by idiosyncratic characteristics of the currencies involved.

The remainder of this paper proceeds as follows. We selectively discuss earlier literature in [Section 2](#). [Section 3](#) details our data and portfolio formation procedure. [Section 4](#) describes momentum returns in FX markets and compares momentum strategies with benchmark technical trading rules and the popular carry trade, while [Section 5](#) discusses the results of our tests seeking to explain the high returns to currency momentum strategies. [Section 6](#) provides robustness checks and [Section 7](#) concludes. Additional results can be found in an Appendix to this paper.

² We use the term “limits to arbitrage” here to mean that trading momentum strategies exposes the investor to risks not captured by traditional covariance risk measures so that an anomaly like momentum returns is not easily exploitable. This definition is in line with much of the recent literature but it should be noted that the term (originally due to Keynes) initially referred to the market’s inability to exploit risk-free arbitrage opportunities. Relative to this more precise definition, our tests are more closely related to “limits to speculation”.

2. Related Literature

Academic studies about momentum strategies are mostly focused on stock markets but momentum effects have been also detected in bond and commodity markets. To set the stage, we briefly survey this literature before we turn to FX markets and highlight the contributions of this paper.

Stock market momentum. Momentum effects are well documented in equity markets for almost two decades. The empirical literature is highly influenced by the work of [Jegadeesh and Titman \(1993\)](#), who show in a thorough analysis of the U.S. stock market that simple momentum strategies generate high returns, in the order of about 12% p.a., and are difficult to rationalize by standard asset pricing models. Subsequent studies extend the original research into new domains, including many countries worldwide beyond the U.S. (e.g. [Rouwenhorst, 1998, 1999](#); [Chan, Hameed, and Tong, 2000](#); [Chui, Titman, and Wei, 2010](#)) and higher frequencies ([Gutierrez Jr. and Kelley, 2008](#)).

While equity momentum is an established empirical fact, explanations have been heavily disputed. The major approaches to explain momentum can be classified as (i) risk-based and characteristics-based explanations, (ii) explanations invoking cognitive biases or informational issues, and (iii) explanations based on transaction costs or other forms of limits to arbitrage.

Starting with risk-based and characteristics-based explanations (i), early studies show that momentum returns are difficult to rationalize by covariance risk with standard factors (e.g. [Fama and French, 1996](#); [Jegadeesh and Titman, 2001](#)). In the same vein, linking momentum to macroeconomic risk has proven rather challenging.³ By contrast, firm-specific characteristics have been shown to be linked to momentum, e.g. momentum appears to be stronger among smaller firms ([Hong, Lim, and Stein, 2000](#)), among firms with lower credit rating ([Avramov, Chordia, Jostova, and Philipov, 2007](#)), and among firms with high revenue

³For instance, [Chordia and Shivakumar \(2002\)](#) find support for time-varying risk factors explaining momentum returns, whereas [Griffin and Martin \(2003\)](#) and [Cooper, Gutierrez, and Hameed \(2004\)](#) do not.

growth volatility (Sagi and Seasholes, 2007). Also, momentum returns appear to a large extent concentrated in firms with a high likelihood to go bankrupt (Eisdorfer, 2008). Empirical work invoking behavioral biases (ii) in explaining momentum – focusing for example on investors’ under-reaction to news – also featured prominently since the beginning of the debate (Jegadeesh and Titman, 1993) and in subsequent work (e.g. Jegadeesh and Titman, 2001; Grinblatt and Han, 2005; Hvidkjaer, 2006).⁴ Stressing how information is incorporated into prices, Chan, Jegadeesh, and Lakonishok (1996) provide early evidence that analysts’ earnings forecasts respond gradually to news which might generate under-reaction. Hong, Lim, and Stein (2000) demonstrate in detail the relation between weak analyst coverage and stronger momentum.⁵ A final strand explores the role of transaction costs or limits to arbitrage (iii) in explaining momentum. Lesmond, Schill, and Zhou (2004) state that reasonably high transaction costs may wipe out momentum profits. Korajczyk and Sadka (2004) qualify this finding as they argue that momentum strategies may be designed in a way to limit transaction costs; this will lead to a more moderate cost level so that even very large momentum portfolios (with assets worth more than one billion U.S. dollars) are still highly profitable.

Momentum in bonds and commodities. Momentum has also been shown to exist in other asset classes. Regarding bond markets, momentum strategies do not work for investment-grade bonds (Gebhardt, Hvidkjaer, and Swaminathan, 2005) or bonds at the country level (Asness, Moskowitz, and Pedersen, 2009), but yield positive returns for non-investment grade *corporate* bonds (Jostova, Nikolova, Philipov, and Stahel, 2010). Further analysis shows that momentum returns are not related to liquidity but seem to reflect default risk in the winner and loser portfolios. Regarding commodity markets, the high returns to momentum strategies are shown to be related to market states with low level of inventories that indicate higher risk (Gorton, Hayashi, and Rouwenhorst, 2008). These findings tentatively suggest common sources of momentum profits which seem to be based on the risk

⁴Behavioral models e.g. by Daniel, Hirshleifer, and Subrahmanyam (1998), Barberis, Shleifer, and Vishny (1998), Hong and Stein (1999) account for momentum effects by allowing for deviations from fully rational behavior such as over-confidence, slow updating of investor beliefs and information imperfections.

⁵In addition, analyst behavior will lead, during the period of information incorporation, to information heterogeneity among investors, which is shown by Verardo (2009) to be related to momentum.

characteristics of the underlying assets.

Currency momentum. In contrast to the extensive literature on momentum strategies in stock markets, the literature on currency momentum has mostly developed a somewhat different line of research. The most striking difference is the fact that currency momentum studies generally do not analyze momentum in a cross-section of currencies but in the time-series of single exchange rates, often framed as “technical trading rules”.⁶ This literature is surveyed in [Menkhoff and Taylor \(2007\)](#) and we will discuss it in more depth below. This time-series literature has extensively examined which kinds of trading rules work best.

One exception from the time-series focus is [Okunev and White \(2003\)](#) who analyze a universe of eight currencies over 20 years, from January 1980 to June 2000. At the end of each month, the investor goes long in the currency with the best last-month performance and goes short in the currency with the worst last-month performance. This yields a return of about 6% p.a., which is largely independent of the base currency chosen and of the specific trading rule chosen, i.e. how exactly the best and worst currencies are identified. Thus, there is clear indication that currency momentum strategies may be profitable and thus worthy of a thorough examination.⁷ [Burnside, Eichenbaum, and Rebelo \(2011\)](#) investigate returns to an equally-weighted momentum portfolio that aggregates over momentum positions in individual currencies. They find (as we do in this paper) that standard risk factors cannot account for currency momentum returns.

Technical trading in FX markets. Technical trading in FX is in most cases the same as trend following, that is exploiting the momentum of a market. These time-series momentum strategies include filter rules and moving average rules. A filter rule gives the signal to invest (to take a short position) in a currency if a defined upwards (downwards) exchange rate change has occurred, such as a 1 or 2 percent change. A moving average rule gives signals if short-

⁶See, e.g., [Harris and Yilmaz \(2009\)](#), [Neely, Weller, and Ulrich \(2009\)](#), and [Serban \(2010\)](#) in this respect.

⁷More recently, [Asness, Moskowitz, and Pedersen \(2009\)](#) have also investigated returns to a currency momentum strategy based on ten currencies. The focus of their paper is very different from ours, however, with its primary objective being to explore the commonality of momentum across asset classes.

term exchange rate averages become larger or smaller than longer-term averages.⁸ Simple trend following trading strategies of this kind provide attractive returns, even considering interest rate differentials and transaction costs, as for example the early studies of Dooley and Shafer (1976) or Sweeney (1986) have demonstrated.⁹

These early studies have been challenged by subsequent work examining whether trend following trading strategies are also profitable in later periods. Whereas Dooley and Shafer (1983) and Levich and Thomas III (1993) confirm profitability out-of-sample, studies also covering the 1990s and 2000s find that the above mentioned simple trend following strategies applied to the same set of exchange rates no longer yield attractive returns (see, e.g., Olson, 2004; Pukthuanthong-Le, Levich, and Thomas III, 2007; Neely, Weller, and Ulrich, 2009). However, profits are still found if either new forms of trend following strategies or new exchange rates are considered.¹⁰

Contributions of this paper. In contrast to the abundance of time-series studies, there is little evidence on cross-sectional aspects of currency momentum, whose importance has clearly risen in face of the realities of today's FX markets. Whereas there were about ten convertible and liquid currencies in the 1970s, there are more than 30 currencies available to investors today. And while transaction volumes used to be dominated by banks' FX traders, asset managers of various kinds (including hedge funds) have emerged as some of the key players in today's FX markets. Overall, volumes, tradable assets and participants have changed, which culminates in the perception of FX as a separate asset class, in parallel to e.g. equities and bonds (King, Osler, and Rime, 2011). Even retail investors nowadays

⁸For example, a 1,5 (or 5,20) rule suggests to buy Euro against US-dollar, if the 1- (5-) day US-dollar/Euro rate is higher than its 5-day (20-day) average.

⁹These strategies are also implemented in practice and the widespread use has led, e.g., Lequeux and Acar (1998) to build an index based on moving average rules to serve as a benchmark for Commodity Trading Advisors.

¹⁰Less well-known and less studied forms include channel rules, genetic programming-based rules, Markov model-based rules and others (e.g. Neely, Weller, and Dittmar, 1997). Neely and Weller (2011) provide a recent overview of different trading rules in currency trading. Neely, Weller, and Ulrich (2009) show that these rules are still profitable until the end of their sample period in 2005. Pukthuanthong-Le and Thomas III (2008) confirm that standard trading rules in the main exchange rates do not generate profits when recent data are considered, whereas the same rules yield high returns in emerging markets' exchange rates.

have access to various FX investment strategies via structured products. This naturally leads to studying cross-sectional currency momentum taking into account these new features and industry practices.¹¹

In this paper, we go beyond earlier research in a number of directions. First, we analyze a much longer time span and, more importantly, a much larger cross-section of currencies which includes currencies of developed and emerging countries. This extended sample across time and currencies is crucial for our analysis of returns to currency momentum strategies since it allows us to better identify return variation over time (and, hence, states of the business cycle) as well as across currencies that are structurally different and should have different exposures to global risk factors. Second, we can take explicit account of transaction costs, which is crucial since momentum returns are only relevant as long as they survive realistic transaction costs. Third, we take a close look at possible limits to arbitrage (which are a key theme in the recent literature on equity momentum) and investigate the role of idiosyncratic return volatility, country risk, and the risk of exchange rate stability. In sum, we provide a detailed account of the economic anatomy and drivers of currency momentum strategies that has been missing in the literature until now.

3. Data and Currency Portfolios

This section describes our data, the computation of currency excess returns, and the construction of momentum portfolios.

Data source and sample currencies. The data for spot exchange rates and 1-month forward exchange rates cover the sample period from January 1976 to January 2010, and are obtained from BBI and Reuters (via Datastream). We denote the spot and forward rates in logs as s and f , respectively. Spot and forward rates are end-of-month data (last trading day

¹¹We thank an anonymous referee for pointing this out. An investment product such as the Currency Momentum ETF of Deutsche Bank, which is accessible even for retail investors, may serve as an example of these new trends.

in a given month) and are therefore not averaged over a month.

Our total sample consists of the following 48 countries: Australia, Austria, Belgium, Brazil, Bulgaria, Canada, Croatia, Cyprus, Czech Republic, Denmark, Egypt, Euro area, Finland, France, Germany, Greece, Hong Kong, Hungary, India, Indonesia, Ireland, Israel, Italy, Iceland, Japan, Kuwait, Malaysia, Mexico, Netherlands, New Zealand, Norway, Philippines, Poland, Portugal, Russia, Saudi Arabia, Singapore, Slovakia, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, Taiwan, Thailand, Ukraine, United Kingdom.

It is worth noting that, compared to e.g. [Lustig, Roussanov, and Verdelhan \(2011\)](#) or [Menkhoff, Sarno, Schmeling, and Schrimpf \(2011\)](#), whose samples start in 1983 and have seven currency pairs in the beginning of the sample (mainly) based on BBI data quoted against the U.S. dollar, we employ a longer time series that extends back to 1976. We do so by complementing BBI data (which only start in 1983) with Reuters data quoted against the British Pound as in [Burnside, Eichenbaum, Kleshchelski, and Rebelo \(2011\)](#). We have a total of 16 currencies for this longer time span and convert these data to quotations against the U.S. dollar. These 16 countries are: Austria, Belgium, Canada, Denmark, France, Germany, Ireland, Italy, Japan, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, and the United Kingdom. In addition to the larger cross-section and longer time series, we also have bid and ask quotes for spot and forward rates available so that we can adjust for transaction costs for the whole period from 1976 to 2010.

Finally, we note that our effective sample size varies over time as data for emerging countries become available or when currencies cease to exist, e.g., due to the adoption of the Euro. To illustrate this point we plot the number of currencies with available data for each month of our sample in Figure 1 (solid line). As can be inferred from this graph, our sample does not cover all 48 currencies at the same time since data availability varies naturally due to inclusion and exclusion of currencies. The total sum of actual observations (currency-month combinations) is 9,403 as opposed to the theoretical maximum of 19,584 ($408 \text{ months} \times 48 \text{ currencies}$). Individual start and end dates for each currency are shown in Table A.1 in the Appendix.

FIGURE 1 ABOUT HERE

The dashed line in Figure 1 also shows the number of available currencies which are not tightly pegged to other currencies. As may be expected, there are fewer currencies of this sort, especially in the very early part of the sample. While it is not problematic per se to perform momentum trading strategies in tightly linked currencies, one would expect that momentum profits should be relatively lower in the very early years of our sample. This is what we find in our empirical analysis below.¹²

Currency excess returns. Excess monthly returns to a U.S. investor for holding foreign currency k are given by

$$rx_{t+1}^k \equiv i_t^k - i_t - \Delta s_{t+1}^k \approx f_t^k - s_{t+1}^k \quad (1)$$

where s and f denote the (log) spot and 1-month forward rate (foreign currency unit per USD), respectively. Δs denotes the log spot rate change or return. Descriptive statistics for excess returns, forward discounts, and bid-ask spreads are reported in the Appendix (Table A.1).

For future reference, we also define net currency excess returns, i.e. currency excess returns after bid-ask spreads. These returns only apply when investigating dynamic investment strategies (momentum strategies in our case), where investors form portfolios of currencies. We detail the construction of portfolios below and simply define how we adjust for transaction costs here.

The net return for a currency that enters a portfolio at time t and exits the portfolio at the end of the month is computed as $rx_{t+1}^l = f_t^b - s_{t+1}^a$ for a long position and $rx_{t+1}^s = -f_t^a + s_{t+1}^b$ for a short position. An a (b) superscript indicates the ask (bid) quote. A currency that enters a portfolio but stays in the portfolio at the end of the month has a net excess return

¹²We thank an anonymous referee for pointing this out.

$rx_{t+1}^l = f_t^b - s_{t+1}$ for a long position and $rx_{t+1}^s = -f_t^a + s_{t+1}$ for a short position, whereas a currency that exits a portfolio at the end of month t but already was in the current portfolio the month before ($t - 1$) has an excess return of $rx_{t+1}^l = f_t^b - s_{t+1}^a$ for a long position and $rx_{t+1}^s = -f_t^a + s_{t+1}^b$ for a short position. Hence, since forward contracts in our sample have a maturity of one month, the investor always incurs transaction costs in the forward leg of his position but does not always have to trade the spot market leg of his position if he stays invested in a foreign currency. In addition, we assume that the investor has to establish a new position in each single currency in the first month (January 1976) and that he has to sell all positions in the last month (at the end of January 2010). Note that bid and ask rates are daily (not averaged over the month) so that they correspond exactly to the end-of-month data for spot and forward rates.

However, one has to bear in mind that bid-ask spreads from BBI/Reuters are based on indicative quotes which are “too high” (see e.g. Lyons, 2001) relative to actual effective spreads in FX markets so that our results with net returns (after deducting the bid-ask spread) should be understood as undercutting the lower bound on the profitability of momentum strategies and not as the “exact” return. For this reason, we frequently provide results with and without transaction costs below in our empirical analysis. We denote returns or spot rate changes after deducting bid-ask spreads as “net returns” and “net spot rate changes”, respectively.

Portfolio construction. At the end of each month, we form six portfolios based on lagged returns over the previous $f = 1, 3, 6, 9, 12$ months (f denotes the formation period) and these portfolios are held for $h = 1, 3, 6, 9, 12$ months (h denotes the holding period). The one sixth of all available currencies in a given month which have the lowest lagged returns are allocated to the first portfolio (denoted “Low”), the next sixth is allocated to portfolio 2, and so on, and the one sixth of all currencies with the highest lagged returns are allocated to the sixth portfolio (denoted “High”). Hence, this procedure yields a time-series of six currency momentum portfolios’ excess returns and is analogous to the construction of momentum

portfolios in the equity market literature.¹³

However, since interest rate differentials (forward discounts) contribute a significant share of the excess return of currency investments, we also track the pure spot rate changes of momentum portfolios themselves and report them separately in many tables. This way, we can check whether currency momentum is mainly driven by interest rate differentials or whether it occurs in spot rates, too.

Finally, in most analyses we work with the portfolio which is long in the winner currencies (portfolio “High”) and short in the loser currencies (portfolio “Low”). These portfolios are denoted $MOM_{f,h}$ where f and h represent the formation and holding period, respectively, as defined above. We also refer to these portfolios simply as “long-short” momentum portfolios or “high minus low” portfolios. An important feature of these long-short portfolios is that they are dollar-neutral, since the dollar component cancels out when taking the difference between (any) two portfolios.

4. Characterizing Currency Momentum Returns

In this section, we present our main empirical results regarding the profitability and characteristics of currency momentum strategies (Section 4.1), the stability of the strategies out-of-sample (Section 4.2), the difference between currency momentum and technical trading rules (Section 4.3), the difference between currency momentum and carry trades (Section 4.4), and the long-run return behavior of momentum strategies (Section 4.5).

¹³Lustig and Verdelhan (2007) were the first to form portfolios of currency excess returns to be able to explain returns to the carry trade. This approach of forming currency portfolios has proved very useful in uncovering the economic drivers of carry trade risk premia and has been followed by several other papers afterwards. This way of constructing momentum returns differs from much of the earlier literature on technical trading in currency markets which mostly works in the time-series of individual currency pairs (and then potentially aggregate across all currencies in the sample). Our approach is closer to how momentum is studied in the equity market literature and it is also closely related to how the financial industry sets up tradable momentum portfolios. For example, Deutsche Bank offers a currency momentum ETF based on G10 currencies and the underlying index is long (short) in the three best (worst) performing currencies over the last twelve months (Deutsche Bank, 2010).

4.1. Returns to Momentum Strategies in Currency Markets

Table 1, Panel A, shows average annualized excess returns (left panel) and spot rate changes (right panel) for a number of high minus low momentum portfolios with formation and holding periods each varying between one and twelve months: $f, h = 1, 3, 6, 9, 12$. Average excess returns in the left panel are based on sorting on lagged excess returns, and average spot rate changes in the right panel are based on sorting on lagged spot rate changes. To provide a perspective on profitability of FX momentum relative to risk, Panel B of Table 1 reports Sharpe ratios for the same strategies.

Turning to excess returns in the left panel first, we find that momentum strategies yield substantial (and statistically highly significant) excess returns of about 6 – 10% for short holding periods of one month and their profits slowly fade out when increasing the holding period. The latter finding is quite pronounced since there is a monotone decline in average excess returns when moving from short holding periods to longer holding periods h for a given formation period f . However, we find many instances of significant momentum returns for strategies with longer holding periods as well, so that momentum is not confined to very short holding periods.

In the right panel of Table 1, Panel A, we also report the average difference between spot rate changes for the high and low portfolio. For ease of exposition, we actually report the negative of the log spot rate change (in the notation of Section 3) so that higher values indicate a positive contribution of spot rate movements to a momentum strategy's total excess return. Interestingly, the profitability of currency momentum strategies is also clearly visible in spot rate changes themselves and is thus not mostly driven by the interest rate differential as is the case for carry trades (see, e.g., Lustig, Roussanov, and Verdelhan, 2011). In fact, the strategy with a twelve months formation period is completely driven by favorable spot rate changes and the interest rate differential reduces the excess return somewhat.

TABLE 1 ABOUT HERE

As noted above, results tend to be strongest for a holding period of $h = 1$ month. We therefore focus on these strategies in most of the following analysis as they seem to present the hardest challenge when trying to understand momentum returns in currency markets. Since the level of average excess returns is also clearly dependent on the formation period f , we provide results for the three strategies with $f = 1, 6$, and 12 months in our empirical analyses below. In sum, most of our analysis in the remainder of the paper focuses on the three benchmark strategies $MOM_{1,1}$, $MOM_{6,1}$, and $MOM_{12,1}$.¹⁴

As a first and simple means of investigating a possible link between momentum returns and the state of the business cycle, and to provide a graphical exposition of momentum returns accruing to investors, Figure 2 shows cumulative excess returns for the three benchmark momentum strategies $MOM_{1,1}$, $MOM_{6,1}$, and $MOM_{12,1}$ over the full sample period. Shaded areas correspond to NBER recessions. As illustrated by the figure, there is no obvious correlation of momentum returns with the state of the business cycle (as examined later in Section 5.2). However, the three benchmark momentum strategies show some co-movement but are not perfectly correlated.

FIGURE 2 ABOUT HERE

Sharpe Ratios. In order to get a first measure of risk-adjusted returns, Panel B of Table 1 presents Sharpe Ratios for the momentum strategies shown in Table 1 above in Panel A, and “normalized spot rate changes” (average spot rate changes divided by their standard deviation) in Panel B. Corroborating the evidence above, currency momentum strategies seem highly profitable, at least for a subset of strategies. For example, the annualized Sharpe Ratio of the MOM(1,1) strategy is 0.95, which seems very high, even in comparison to carry trades (see, e.g., Menkhoff, Sarno, Schmeling, and Schrimpf, 2011, who report an annualized Sharpe Ratio of 0.82 for a carry trade strategy). Hence, even when taking risk into account on the basis of Sharpe Ratios, momentum strategies seem highly attractive. In addition, we

¹⁴One might worry that some currencies were not always tradable during our sample period due to, e.g., capital account restrictions. We provide robustness checks on this issue in Section 6.1.

see from Panel B of Table 1 that this performance is largely driven by spot rate changes and that it is not dominated by the interest rate component of excess returns.

Momentum returns and size of the cross-section. As noted above in the previous section, our effective sample size never exceeds 40 currencies and is therefore relatively small compared to sample sizes used in, e.g., the equity momentum literature. However, it is well known from earlier work that even small portfolios of currencies can yield large gains from diversification since currencies tend to be less correlated than stocks (e.g., [Burnside, Eichenbaum, and Rebelo, 2008](#)). In order to explore the link between the size of the cross-section and the magnitude of momentum returns, we conduct a stylized simulation experiment as follows. In each run i , we randomly draw (without replacement) a set of N currencies from the set of all 48 currencies while imposing the restriction that we have data for at least six currencies in each month of the sample period from January 1976 to January 2010. We then calculate average annualized momentum excess returns for a MOM(1,1) strategy and save this result. We do this 5,000 times for each cross-section size N and average over momentum profits to obtain an estimate of the “typical” momentum profit conditional on observing a cross-section of size N . For $N = 48$ we simply report the momentum profit from Table 1.

Figure 3 shows results from this exercise and it can be seen that expanding the size of the cross-section is very useful for small cross-sections but much less important for larger cross-sections. In other words, there are decreasing gains from expanding the size of the tradable currency universe. The maximum level of returns is roughly obtained for a cross-section of size $N = 36$. Hence, although our cross-section is far from what is used in the equity market literature, one can be confident that the results are quite representative of the currency market as a whole.

FIGURE 3 ABOUT HERE

4.2. Out-of-sample Perspective

Our setup to illustrate FX momentum profits, which is akin to the equity literature, has a clear out-of-sample component, since we form portfolios based on lagged information only. Hence, the momentum strategies discussed above are implementable in real time. However, average returns can vary markedly across different strategies (that is, different combinations of formation and holding period), and can also be fairly low. For example, the strategy with a twelve months formation and holding period only yields 1.89% p.a. over the full sample whereas the strategy with a one month formation and holding periods experienced an annualized average return close to 10%. This particular information is only available ex post and an investor could not have conditioned on this information in 1976. Hence, it is interesting to examine whether investors could have actually exploited these momentum profits taking into account that there is ex ante uncertainty about which specific momentum strategy to follow.¹⁵ Put differently, do specific momentum strategies identified to be attractive in-sample continue to do well?

We tackle this question by investigating returns to what we term “Momentum²” strategies. To do so, we imagine an investor who can invest in 144 different strategies (all combinations of $f = 1, 2, \dots, 12$ and $h = 1, 2, \dots, 12$) and has to rely on some mechanism to select between these different strategies. A natural mechanism in our context is to let the investor rely on momentum in lagged momentum returns (as measured over an evaluation period). More specifically, we form 9 portfolios out of the universe of 144 possible momentum strategies. These nine portfolios are based on a ranking of the momentum strategies themselves by their lagged returns during an evaluation period, hence the term Momentum². Results for this exercise are shown in Table 2 which reports returns for all 9 Momentum² portfolios (from “worst” lagged returns to “best” lagged returns) and a “best minus worst” portfolio. For robustness we show results for lags of 1, 3, 6, 9, 12, 60, and 120 months over which individual momentum strategies are evaluated. As can be seen, using lagged momentum returns to identify future momentum returns seems feasible. For example, conditioning just on last

¹⁵Silber (1994) also investigates whether trading strategies identified as profitable over an in-sample period continue to perform well in an out-of-sample period.

month's return across all possible strategies leads to an annualized average excess return of 7.67% p.a. As for the simple momentum strategies above, we see a declining pattern in returns when moving to longer selection windows. For example, using a window of 120 months leads to much lower returns of only 2.70% p.a., which, however, are still significantly different from zero. Most importantly, these results indicate, however, that specific FX momentum strategies that performed well in the past tend to continue to do well and are thus quite stable.

TABLE 2 ABOUT HERE

While the above analysis confirms that momentum returns are exploitable in an out-of-sample setting, we further examine this issue from a somewhat different angle by a simple investigation of the sub-sample stability of momentum profits. To do so, Table A.2 in the Appendix shows average annualized excess returns and Sharpe Ratios for four subperiods of equal length. We report results for formation periods of $f = 1, 3, 6, 9, 12$ and a holding period of one month. As can be seen, the ranking of these five different strategies is fairly stable over the four subperiods. In other words, it is never the case that one strategy does extremely well in one subperiod but then produces large losses in the next subperiod. Overall, we conclude that it should have been possible for an investor to exploit momentum strategies in real time.

4.3. *Comparing Momentum and Technical Trading Rules*

The results presented above suggest that momentum effects in the cross-section of currencies are quite strong and that momentum strategies consequently yield high excess returns and Sharpe Ratios. However, an important question is whether the currency momentum returns documented above can be regarded as a novel phenomenon per se or whether they may merely reflect returns to technical trading strategies that have been documented extensively in the earlier literature.

To investigate this issue, we compute returns to three benchmark moving average cross-over

rules that have been employed frequently in earlier work on technical trading in FX markets. These strategies are based on moving averages of 1 and 20 days (1, 20), 1 and 50 days (1, 50), and 1 and 200 days (1, 200) (see, among others, Dooley and Shafer, 1983; Levich and Thomas III, 1993; Neely, Weller, and Ulrich, 2009).¹⁶ While it is clearly not the case that these three strategies are perfect proxies for *all* possible technical trading strategies, their prominence in the earlier literature makes them interesting for comparison with our cross-sectional currency momentum strategies.

To set the stage, we first compute returns to these moving average rules for all currencies in our sample individually and then aggregate these strategies into an equally-weighted portfolio. Panel A of Table 3 reports descriptive statistics for the three rules, which show that these strategies are profitable, with annual mean excess returns around 5% and high annual Sharpe Ratios between 0.77 and 0.88. Hence, these strategies form an interesting benchmark for our momentum returns.

To assess whether returns to the moving average rules described above capture returns to the currency momentum strategies, we run regressions of momentum returns for the MOM(1,1), MOM(6,1), and MOM(12,1) strategies on returns of the three moving average rules. Results are shown in Panel B of Table 3.

TABLE 3 ABOUT HERE

It can be seen that, even though moving average rule returns and currency momentum are to some extent correlated, the largest R^2 only amounts to 26%. More importantly, all intercept estimates (α s) are large in economic terms and strongly significant in statistical terms. Hence, it seems fair to conclude that currency momentum is not closely related to benchmark technical trading strategies as studied in the earlier literature, and that controlling for returns of these trading rules does not wipe out returns to our cross-sectional currency momentum.

¹⁶These trading strategies generate a buy (sell) signal, when the shorter moving average crosses the longer moving average from below (above).

In addition, we also examine returns to individual currencies' momentum strategies, i.e. where an investor is long or short in each currency depending on lagged returns in the same currency (this strategy is also studied in [Moskowitz, Ooi, and Pedersen, 2011](#)). We report descriptive statistics for returns of each currency in Panel A of Table [A.3](#) in the Appendix, along with the average across countries, an equally weighted portfolio of all individual currencies' strategies, and, for comparison, the cross-sectional momentum strategy employed in this paper in Panel B. It can be seen that most of these time-series momentum strategies are profitable on average (Panel A of Table [A.3](#)) but that an aggregate strategy (the equally weighted portfolio, EW, in Panel B) is less profitable than a cross-sectional momentum strategy (MOM(1,1) in Panel B), which has a much higher average excess return (almost twice as high) and Sharpe Ratio.

4.4. *Comparing Currency Momentum and the Carry Trade*

An important question is to what extent momentum strategies simply capture the same information as the popular carry trade strategy in FX markets, where investors go long in high interest rate currencies and short in low interest rate currencies. After all, interest rate differentials are strongly autocorrelated and spot rate changes do not seem to adjust to compensate for this interest rate differential, which is well-known in the literature as the “forward premium puzzle” ([Fama, 1984](#)). Hence, it may be the case that lagged high returns simply proxy for lagged high interest rate differentials and that, therefore, currency momentum returns are very similar to carry trade returns. In order to address this concern, we perform a comprehensive comparison between momentum returns and carry trade returns in this section. The results clearly show that carry trade and momentum strategies, as well as their associated returns, are in fact very different.

Comparing portfolio properties. We first investigate characteristics of momentum and carry trade portfolios, which are reported in Table [4](#). The table shows descriptive statistics for the six momentum portfolios with a formation and holding period of one month and six carry

trade portfolios where currencies are sorted into portfolios depending on their lagged interest rate, as in e.g. [Lustig, Roussanov, and Verdelhan \(2011\)](#) or [Menkhoff, Sarno, Schmeling, and Schrimpf \(2011\)](#).¹⁷

TABLE 4 ABOUT HERE

As can be inferred from this table, there is a monotonically increasing pattern in average returns for both cross-sections but no clear pattern in higher moments of the return distribution. While the level of average returns and standard deviations of the high minus low momentum and carry trade portfolios is roughly similar, we find that the two long-short portfolios are clearly different in terms of their skewness. While the carry trade produces negatively skewed excess returns (also see [Brunnermeier, Nagel, and Pedersen, 2009](#)), we find a slightly positive skewness for the momentum strategy.

More interestingly, the last two rows of each panel show lagged average returns and lagged average forward discounts for each portfolio at the time of portfolio formation. Momentum portfolios do have a positive spread in forward discounts and carry trade portfolios have a positive spread in lagged returns, but these spreads are much lower in absolute value than the spread in the characteristic used for sorting currencies into portfolios. More specifically, the average cross-sectional spread in forward discounts (in annualized terms) at the time of portfolio formation is about 4.6% (5.13% versus 0.44%) for the momentum cross section but averages more than 15% for the carry trade cross section. Similarly, the average spread in lagged returns is almost 6% for the momentum portfolios (2.94% versus -2.93%) but only 0.84% for the carry trade cross-section. Hence, momentum and carry trade strategies may be somewhat related but are far from being identical.

Return correlations. Table 5, Panel A, shows correlation coefficients between returns to momentum portfolios and carry trade portfolios. We show results for the long-short

¹⁷To conserve space in this table, we focus on the momentum strategy with $f = 1$ and $h = 1$. Results are similar for the other strategies.

momentum strategies $MOM_{1,1}$, $MOM_{6,1}$, and $MOM_{12,1}$ and always report the correlation between corresponding portfolios; e.g. the correlation of momentum portfolio 2 and carry trade portfolio 2, or the correlation between the high minus low (H-L) carry trade and momentum portfolios. It can be seen that the correlations of excess returns for the six portfolios are rather high but that there is basically no correlation between the high minus low portfolios, and the latter represent the way carry trade and momentum strategies are typically implemented by market participants. Thus, the return to following a currency momentum strategy is basically uncorrelated with carry trade returns and this finding holds true regardless of the respective formation period underlying a momentum strategy.

TABLE 5 ABOUT HERE

In contrast, we show in Panel B that the high minus low portfolios of the three momentum strategies are much more highly correlated and reach correlations of more than 70% for $MOM_{6,1}$ and $MOM_{12,1}$. Hence, it seems fair to conclude that returns to different momentum strategies are likely to share a strong common component.

That excess returns to carry trades and momentum strategies are basically uncorrelated in FX markets appears in line with real-world strategies of many currency investors who combine momentum and carry trade positions in their portfolios to take advantage of an alleged diversification benefit from following the two strategies simultaneously.¹⁸ For example, during the recent financial crisis from July 2007 to June 2009, the benchmark momentum strategy with $h = f = 1$ experienced an average monthly return of 0.80% whereas the carry trade yielded a negative average monthly return of -0.05% . The return correlation of these two strategies was as low as -31% over these two years. Hence, the two strategies showed a clearly different behavior during this period.

¹⁸Patton and Ramadorai (2011) for example show in a general universe of hedge funds (not necessarily currency funds) that there is significant exposure to carry trade and momentum-type returns and that this exposure is time-varying. Pojarliev and Levich (2010) show via style regressions that currency fund managers engage in both carry trade and momentum-type strategies. Melvin and Shand (2011) show that currency managers follow momentum strategies but that their exposure to momentum and the way momentum strategies are implemented change over time.

Double sorts. Next, we provide results based on double sorts. To this end, we first double-sort currencies into two portfolios depending on whether a currency has a lagged forward discount above or below the median (of all available currencies), and then into three portfolios depending on their lagged excess return. Portfolios are re-balanced each month (i.e. $h = 1$). Table 6 shows results for these double sorts for formation periods of $f = 1, 3, 12$ months. There is no material difference between momentum returns among high versus low interest rate currencies. For example, the high minus low momentum return for a strategy with a one month formation period based on low interest rate currencies is 5.06% p.a. on average, whereas it is 5.36% p.a. for high interest rate currencies. Hence, the difference between these two high minus low momentum portfolios is less than 0.30% p.a. and not statistically significant (with a t -statistic of only 0.17). Findings for the other two formation periods are very similar.

TABLE 6 ABOUT HERE

As above, we do not find a strong relation between momentum and carry trade strategies and the double sorts suggest that the two strategies are largely independent. In fact, going long in currencies with high lagged returns and high interest rates whilst shorting currencies with low returns and low interest rates generates an excess return of 10.52% p.a., which is even larger than the spread in both momentum or carry trade portfolios taken individually.

Cross-sectional regressions. Finally, we want to separate the effects of lagged excess returns and lagged interest rate differentials on future excess returns. To this end, we run Fama-MacBeth type cross-sectional regressions of currency excess returns (or spot rate changes) on (i) lagged excess returns over the last l months, (ii) lagged forward discounts, and/or (iii) lagged spot rate changes for each month of our sample, i.e.

$$rx_t^k = \alpha_t + \beta_{rx,t} rx_{t-\ell;t-1}^k + \beta_{FD,t} (f_{t-1} - s_{t-1}) + \beta_{\Delta s,t} \Delta s_{t-\ell;t-1}^k + \varepsilon_t \quad (2)$$

where the subscript $t - \ell; t - 1$ refers to a variable defined over the last ℓ months using information available at time $t - 1$. This procedure yields a time-series of coefficient estimates (α_t, β_t) and we report the mean of these time series and t -statistics based on [Newey and West \(1987\)](#) standard errors in Table 7 in the spirit of the approach by [Fama and MacBeth \(1973\)](#).¹⁹

These cross-sectional regressions serve to disentangle the information contained in lagged returns (or spot rate changes) and forward discounts for future excess returns (or spot rate changes) in a regression framework and on the level of individual currencies. Momentum strategies require individual currencies' excess returns to vary cross-sectionally in a way that is predictable by lagged returns. Cross-sectional regressions allow us to test for this effect while simultaneously controlling for interest rate differentials and, hence, complement the double sorts above which work on a portfolio level and do not necessarily control for both factors at the same time due to sequential sorting.

Panel A shows results for regressions where we use lagged excess returns, forward discounts, and/or spot rate changes over the last month as explanatory variables, whereas Panels B and C show results for values of l equal to six and twelve, respectively.²⁰

Turning to results for excess returns first (left part of Table 7), we find that lagged returns, lagged forward discounts, as well as lagged spot rate changes are cross-sectionally positively related to subsequent currency returns even when including them in joint specifications. Hence, momentum effects are robust to controlling for forward discounts (interest rate differentials). Furthermore, it is noteworthy that lagged spot rate changes do about as well as lagged excess returns in the cross-sectional regressions so that momentum seems to originate from spot rate changes and not from lagged interest rate differentials, which corroborates our finding that carry trades and momentum are different.

TABLE 7 ABOUT HERE

¹⁹See for example [Gutierrez Jr. and Kelley \(2008\)](#), who employ a similar methodology.

²⁰For ease of interpretation, we multiply spot rate changes by minus one, so that higher values mean that the foreign currency is appreciating against the USD.

The right part of Table 7 shows the same calculations but with spot rate changes as dependent variables. While the effect of lagged returns or spot rate changes is very similar to our results described above, we find that the forward discount has a *negative* impact on future spot rate changes. However, the coefficients based on univariate regressions are always smaller than one in absolute value. Hence, a one percent higher interest rate in a foreign country is only followed by a depreciation smaller than one percent relative to other currencies' excess returns against the USD, consistent with the existence of a forward bias (Fama, 1984). Note that these are *cross-sectional* regressions so that results do not necessarily translate into a time-series setting in which the forward premium puzzle has typically been studied.

4.5. *Post-formation Momentum Returns*

Jegadeesh and Titman (2001) suggest that momentum returns are driven by slow information diffusion that leads to under-reaction and persistence in returns (also see Chui, Titman, and Wei, 2010). This initial under-reaction may furthermore be accompanied by subsequent over-reaction which magnifies the drift in returns but has to be corrected over the long run. To investigate these issues, Jegadeesh and Titman (2001) study the post-formation holding period returns of momentum strategies over longer time spans (i.e. the returns over long horizons after portfolio formation where the portfolio composition is held constant). They find a (roughly) “inverted U-shaped pattern”, i.e. returns tend to increase for several months up to one year after portfolio formation but then peak and start to decrease significantly. Jegadeesh and Titman interpret this pattern as evidence of initial under-reaction which drives prices and subsequent over-reaction to the series of high returns, pushing prices up above the fundamental value of the asset. This over-reaction is then corrected over longer periods, leading to the observed predictable pattern of increasing and decreasing returns after portfolio

formation.²¹

As a first check of this hypothesis for currency markets, we plot cumulative post-formation excess returns over periods of $1, 2, \dots, 60$ months for the zero-cost long-short momentum portfolios with a one, six, and twelve months formation period (i.e. MOM_1, MOM_6 , and MOM_{12}) in Figure 4. Returns in the post-formation period are overlapping since we form new portfolios each month but track these portfolios for 60 months. There is a clear pattern of increasing returns which peaks after 8 – 12 months across strategies and a subsequent period of declining excess returns. The decline is more pronounced for momentum strategies with longer formation periods. Thus, on the face of it, this evidence looks very similar to the pattern identified in equity markets as in Jegadeesh and Titman (2001). This result is interesting since it suggests that currency and equity market momentum may have similar origins.²²

FIGURE 4 ABOUT HERE

In sum, these results on currency momentum are consistent with those on stock market momentum, where momentum returns may be (at least partly) driven by slow information processing and investor over-reaction. However, given the highly liquid FX market which is dominated by professional traders and investors it is hard to believe that investor irrationalities of this kind are not quickly arbitrated away. Thus, it is worthwhile to examine possible limits to arbitrage activity which could explain the persistence of momentum profits in FX markets. This is addressed in the next section.

²¹There is relatively little work on behavioral effects in currency markets (compared to equity markets). Burnside, Han, Hirshleifer, and Wang (2010) recently show, however, that concepts from behavioral finance may be useful to understand FX phenomena as well. In addition, Bacchetta and van Wincoop (2010) argue that many FX portfolios are still not actively managed but that portfolio decisions are often taken infrequently, which can be fully rational due to the costs of portfolio adjustments. This mechanism could also account for slow diffusion of information into prices in FX markets. Investors' infrequent portfolio adjustment decisions, slow-moving capital deployed to exploit arbitrage opportunities and the implications of these aspects for the dynamics of asset price movements are also demonstrated recently in Duffie (2010).

²²We also provide the same results for post-formation drift in cumulative spot rate changes in Figure A.1 in the Appendix and find a very similar pattern (although with a somewhat lower magnitude with respect to the initial price increases) so that the result discussed above does not seem to be driven by interest rate differentials but also stems from price changes.

5. Understanding the Results

5.1. Transaction Costs

What role do transaction costs play for momentum returns? To address this question, we first report momentum returns after transaction costs in Table 8, which is otherwise identical to Table 1 but provides an adjustment for transaction costs. For this table, we impose the full quoted bid-ask spreads. This spread is known to be too large relative to actual effective spreads (Lyons, 2001). Hence, these results are likely to underestimate momentum returns (or equivalently to provide a lower bound on profitability), whereas neglecting spreads clearly overstates momentum returns.

TABLE 8 ABOUT HERE

The results show that transaction costs could be an important factor for understanding momentum returns in currency markets (Burnside, Eichenbaum, Kleshchelski, and Rebelo, 2006; Burnside, Eichenbaum, and Rebelo, 2007). When applying the full spread, returns for the best strategy (with $f, h = 1$) drop from nearly 10% to about 4% p.a. and they wipe out most of the profit of many other strategies. Interestingly, the effects of transaction costs on the average spot rate changes of portfolios (which are adjusted for bid-ask in an analogous fashion to excess returns) are relatively less affected. To make the full effect of transaction costs more transparent, we also plot cumulative net excess returns (after transaction costs) for the three baseline strategies $MOM_{1,1}$, $MOM_{6,1}$, and $MOM_{12,1}$ in Figure 5. Again, shaded areas correspond to NBER recessions. It can be seen that FX momentum strategies are much more profitable (after transaction costs) in the later part of the sample, but momentum strategies do not always deliver high returns to investors. Instead, there is much variation in profitability.

FIGURE 5 ABOUT HERE

Next, given that the quoted spread is known to be too high relative to effective spreads, we follow [Goyal and Saretto \(2009\)](#) and report results for momentum excess returns after transaction costs of 75% (Panel A) and 50% (Panel B) of quoted spreads in [Table 9](#).

TABLE 9 ABOUT HERE

Results for these more realistic bid-ask spread adjustments indicate that transaction costs clearly matter but that they are not the sole driver of FX momentum returns as we find that many strategies still yield economically high and statistically significant returns on average.

Further scrutinizing this issue, we can break up the importance of transaction costs into turnover across portfolios and bid-ask spreads across portfolios. We provide results on both issues in the Appendix ([Table A.4](#)). Two main conclusions emerge from this exercise. First, turnover can be extremely high, reaching values of more than 70% per month for the strategy with a one month formation and holding period. Second, the winner and loser currencies do have higher transaction costs than the average exchange rate and the markup ranges from about 2.5 to 7 basis points per month. Accordingly, trading in the winner and loser currencies (as is necessary to set up a momentum strategy) is more costly than trading in the average currency pair. Hence, transaction costs clearly matter to a considerable extent.

However, given that transaction costs should be expected to decline over time due to more efficient trading technologies (such as electronic trading networks operated by e.g. EBS and Reuters), it seems unclear whether transaction costs are able to fully explain momentum returns. [Figure 6](#) shows average bid-ask spreads across currencies for each month in our sample and separately for all countries and for the subsample of 15 developed countries as defined above. While there is a lot of time-series variation in average spreads, it is the case that spreads have trended downwards over our sample period. This downward trend is most clearly seen for the sample of developed countries for which we have almost complete data histories and for which average spreads are not driven by the frequent inclusion of emerging market currencies that induce some large spikes in average spreads when looking at the sample of all countries. Overall, the downward trend in bid-ask spreads seems to suggest

that new technology has swamped the positive effect of volatility on bid-ask spreads. Thus, it is interesting to also investigate momentum strategies over a later part of our sample where bid-ask spreads tend to be lower on average since lower transaction costs could either imply (i) higher momentum returns due to lower trading costs or (ii) lower momentum returns since lower trading costs facilitate more capital being deployed for arbitrage activity.

FIGURE 6 ABOUT HERE

Appendix Table A.11 shows results for the same calculations underlying Table 1 above but we only include the period January 1992 to January 2010 in order to learn about whether the profitability of momentum strategies increases or declines over this recent period of low transaction costs. We find that unadjusted momentum returns reach levels similar to those for the full sample (Panel A) but that transaction cost-adjusted net excess returns (Panel B) are clearly higher and, for example, reach average annualized values of more than 7% for the 1-month strategy $MOM_{1,1}$. Thus, lower bid-ask spreads do not necessarily lead to lower (unadjusted) excess returns, which further indicates that transaction costs are not the sole driving force behind momentum effects. This evidence also indicates that momentum returns are a phenomenon which is still exploitable nowadays.

5.2. *Momentum Returns and Business Cycle Risk*

Table 10, Panel A, shows results from univariate time-series regressions of momentum returns on various risk factors or business cycle state variables (see, e.g., Burnside, Eichenbaum, Kleshchelski, and Rebelo, 2011; Sarno, Schneider, and Wagner, 2011, for similar regressions in the context of currency returns). These factors include macro variables or other risk factors from the earlier literature: “Consumption” stands for real growth in non-durables and services consumption expenditures, “Employment” denotes U.S. total nonfarm employment growth, “ISM” denotes the ISM manufacturing index, “IP” denotes growth in real industrial production, “CPI” denotes the inflation rate, “M2” is the growth in real money balances, “Disp Inc” is growth in real disposable personal income, “TED” denotes the TED spread (the

difference between 3-month interbank rate, Libor and 3-month T-Bill rate), “Term” denotes the term spread (20-year maturity minus 3-month T-Bill rate), HML_{FX} is the return to the carry trade long-short portfolio (Lustig, Roussanov, and Verdelhan, 2011), and VOL_{FX} is a proxy for global FX volatility (Menkhoff, Sarno, Schmeling, and Schrimpf, 2011). We note that the alphas in these regressions cannot be interpreted as a measure of risk-adjusted returns for most specifications since we are mainly employing macro variables or other non-return based factors here. Statistical significance at the 5% level or below is indicated by bold numbers. However, looking across momentum strategies and macro-finance risk factors, there is little evidence that exposure to these factors is able to account for momentum returns. The adjusted R^2 s are generally tiny and most slope coefficients are insignificantly different from zero.²³

TABLE 10 ABOUT HERE

Panel B of Table 10 shows a multivariate regression of momentum returns on the three Fama-French factors augmented by the U.S. stock momentum factor (UMD), and it can again be seen that there is basically no explanatory power. Moreover, the alphas in these regressions (which are annualized and in percentages) can be interpreted as the risk-adjusted performance of momentum returns since the factors are excess returns in this case. Across strategies, the alphas are fairly high, as judged by this particular model for returns. Based on earlier research for the U.S. stock market, this result does not come as a surprise regarding the three Fama-French factors but it seems noteworthy that currency momentum is also unrelated to the UMD factor.²⁴

²³As mentioned earlier, one exception is the momentum strategy with a 12 months formation period and global FX volatility. We find a highly significant slope coefficient here and a positive R^2 . Menkhoff, Sarno, Schmeling, and Schrimpf (2011) show for this momentum strategy that innovations to global FX volatility do indeed capture a large amount of the cross-sectional spread in returns and that volatility risk is significantly priced. However, we do not find that FX volatility risk helps much for understanding momentum returns of the strategies with short formation periods of one month or six months.

²⁴We have also experimented with more elaborate cross-sectional asset pricing tests for both macro factors and return-based factors but, as may be expected on the basis of the time-series results reported in Table 10, did not find any improvement in results.

In sum, there is little evidence that standard business cycle variables or portfolio-based risk factors help to understand momentum returns, i.e. it seems that the latter are largely disconnected from U.S. business cycle risk. This finding squares well with earlier results for U.S. equity momentum, which is hard to explain by relying on its covariance with macro risk factors (e.g. [Griffin and Martin, 2003](#); [Cooper, Gutierrez, and Hameed, 2004](#)).

5.3. *Limits to Arbitrage: Time-variation in Momentum Profitability*

Next, we are interested in the stability of momentum returns over time. Since FX market participants (e.g. proprietary trading desks, asset managers and hedge funds) generally have short investment horizons, time-variation in momentum profits could also represent an important obstacle for taking arbitrage positions in FX markets.

Figure 7 plots average excess returns to the three long-short momentum portfolios $MOM_{1,1}$, $MOM_{6,1}$, and $MOM_{12,1}$ over rolling windows of 36 months. The left part shows unadjusted returns while the right part of the figure shows net excess returns after transaction costs. It can be seen that the profitability of momentum strategies is time-varying and that both adjusted and unadjusted returns appear to be higher over the second part of the sample. In fact, momentum returns for all three strategies have been rather high between 2000 and 2005 reaching levels of monthly net excess returns of about 2% per month.

FIGURE 7 ABOUT HERE

Most importantly, this figure also illustrates that momentum returns are far from being constant even over intermediate time intervals of several years. Hence, an investor seeking to profit from momentum returns has to have a long enough investment horizon. This result seems important, since the bulk of currency speculation is accounted for by professional market participants and proprietary traders who have a rather short horizon over which their performance is evaluated ([Lyons, 2001](#)). Hence, momentum strategies are potentially risky for myopic market participants, so that large time-variation in the performance of momentum

returns may impede arbitrage activity by some of the key FX market players.²⁵

5.4. *Limits to Arbitrage: Idiosyncratic volatility*

Unlike in stock markets, there are no natural short-selling constraints in FX markets. However, in order to conduct arbitrage in currency markets, an investor obviously has to set up positions which he may wish to hedge such that the position becomes a pure bet on return continuation but not on any sort of systematic risk. Hence, we investigate whether momentum returns are different between currencies with high or low idiosyncratic volatility (relative to an FX asset pricing model). Finding that currency momentum is stronger among high idiosyncratic volatility currencies would imply that attempts to arbitrage these momentum returns away may be risky since it will be hard to find a second pair of currencies that can be used as a hedge factor unrelated to simple return continuation.

To this end, Panel A of Table 11 shows results from double-sorting currencies first into two portfolios depending on whether a currency has a lagged idiosyncratic volatility above or below the median (of all available currencies), and then into three portfolios depending on their lagged excess return.²⁶ For all three formation periods we study (i.e. f is either 1, 6, or 12), we find that momentum returns are higher among currencies with high idiosyncratic volatility than among currencies with low idiosyncratic volatility (*IVOL*). The returns differences are quite large in economic terms. For example, sorting on lagged idiosyncratic volatility and lagged one month returns leads to an annualized momentum excess return of 3.97% among currencies with low *IVOL*, whereas a momentum strategy among currencies with high *IVOL* yields an average excess return of 8.09% p.a. Thus, momentum strategies

²⁵The role of frictions (e.g. margin and capital constraints) on the deployment of arbitrage capital to investment opportunities by institutional investors is stressed for instance in recent work by [Mitchell, Pedersen, and Pulvino \(2007\)](#). Excellent recent surveys on limits to arbitrage and slow-moving capital which provide an obstacle to the corrective actions of rational arbitrageurs are provided by [Duffie \(2010\)](#) and [Gromb and Vayanos \(2010\)](#).

²⁶Idiosyncratic volatility for each currency j in month t is computed from a regression of currency returns on a constant, the Dollar risk factor, and the HML_{FX} factor of [Lustig, Roussanov, and Verdelhan \(2011\)](#). Idiosyncratic volatility is then computed as the absolute value of the regression residual in month t . We find similar results to those reported below when we employ the volatility risk factor proposed by [Menkhoff, Sarno, Schmeling, and Schrimpf \(2011\)](#).

are much more profitable among currencies with high idiosyncratic risk.

TABLE 11 ABOUT HERE

5.5. *Limits to Arbitrage: Country Risk*

A natural limit to arbitrage in foreign exchange markets is country risk. Institutional constraints such as country limits, for instance, may prevent position-taking in currencies of high risk countries. Arbitrage activity involving these countries' currencies also exposes investors to the risk of potential sudden capital account restrictions and sharp exchange rate moves. This implies that arbitrage strategies involving these countries' currencies are much more risky compared to those involving currencies of well developed and highly stable countries with low risk ratings.

We now perform the same analysis as above but sort instead on a measure of country risk (*CRISK*) and a measure of exchange rate stability risk (*XSTAB*). These data are based on the International Country Risk Guide (ICRG) database from the Political Risk Services (PRS) group.²⁷ We employ the composite country risk rating (which comprises economic, political, and financial risk of a country) as a general proxy for the riskiness of a given country and exchange rate stability risk as a specific proxy for the risk of sharp currency movements.²⁸ Data for these risk proxies start in January 1985 and we employ the log deviation of the risk rating of a country from the rating of the U.S. as a proxy of relative risk for a U.S. investor.

The setup here is somewhat akin to Avramov, Chordia, Jostova, and Philipov (2007, 2010), who show that U.S. stock momentum is mainly concentrated in high credit risk firms which are illiquid and hard to sell short.²⁹ Hence, credit risk proxies for hurdles to arbitrage activity. In our context, we focus on country risk as a natural proxy for limits to arbitrage in

²⁷These data are quite common as proxies for country risk; see e.g. Bekaert, Harvey, Lundblad, and Siegel (2007), who also use risk indicators from this database.

²⁸The exchange rate stability risk proxy measures the perceived risk of large exchange rate movements in the near future.

²⁹In a similar vein, Jostova, Nikolova, Philipov, and Stahel (2010) show that momentum profits in U.S. corporate bond returns derive solely from long and short positions in non-investment grade bonds.

FX markets. High risk countries are more politically unstable, economically less developed and more volatile so that establishing positions in the associated currencies poses non-trivial threats of sudden capital account restrictions and non-convertibility of currency. In short, arbitrage activity involving these countries' currencies should be clearly more risky compared to well developed and highly stable countries with low risk ratings similar to the U.S.

Panels B and C of Table 11 shows results for double sorts on either country risk or exchange rate stability risk and momentum. Corroborating our earlier findings for idiosyncratic volatility, we find that momentum returns are significantly positive and always larger in high-risk countries than in low-risk countries, where momentum strategies do not yield significant excess returns. Hence, for an investor to profit from currency momentum strategies, it is necessary to operate in markets for currencies of risky countries. This is especially important since, unlike momentum strategies in domestic U.S. stocks, investments in foreign currency are always subject to risks of capital controls and non-convertibility. Therefore, country risk should be an important limit to arbitrage activity in FX markets.

Finally, we examine whether our findings above are driven by country risk being related linearly to the cross-sectional spread in momentum returns and whether momentum is differently affected than carry trades. Table A.15 (which, as an example, is based on the strategy with a one month formation and holding period) in the Appendix shows a clear pattern. Country risk and exchange rate stability risk are high for both winner *and* loser currencies (Panel A) in the momentum strategy. Hence, it is not the case that these risk ratings are simple proxies for interest rate differentials which drive our results. Instead, currency momentum strategies require that an investor has to go long and short in the most risky countries. This is especially true since momentum profits stem from both the long and short side of the position (see Table 4, Panel A) so that it is necessary to set up both positions. Contrary to this, the cross-section of forward discount sorted portfolios which form the basis of the carry trade (Table A.15, Panel B) shows a very different pattern. Country risk is highest for carry trade target currencies (high interest rate currencies) and lowest for carry trade funding currencies (low interest rate currencies). This squares well with the finding that most of the carry trade return comes from the long position of the strategy (4, Panel B). In any case,

these results indicate that country risk has a non-linear impact on the cross-sectional spread in momentum portfolios' returns and, again, that the anatomy of carry trade strategies is very different from currency momentum.

Developed countries. Finally, a shortcut to looking at country risk may also be to define a sample of clearly developed countries that have stable exchange rate regimes and are most liquid. Table A.12 in the Appendix shows results before and after transaction costs similar to those in Table 1 but we limit the cross-section to 15 developed countries.³⁰ It is clear from this table that momentum returns are much smaller and basically non-existent after transaction costs when looking at currencies of developed countries. This finding is interesting since it suggests that the profitability of momentum strategies depends on whether smaller and presumably less liquid currencies are included in the investment universe or not. Again, this shows that limits to arbitrage are an important factor in explaining the persistence of momentum returns in FX markets.

6. Robustness and additional tests

6.1. *Capital account restrictions and tradability*

We have documented above that momentum returns are large in FX markets when examining a broad cross-section that also includes smaller currencies from emerging markets. A potential concern regarding these results is whether all currencies have actually been tradable throughout the sample period as there may be capital controls for some countries or other issues rendering trading in these currencies infeasible. Many of these smaller currencies do indeed show up in the loser and winner portfolios quite frequently which is shown in Table A.5 in the Appendix. This table reports the frequency with which each currency is included in the winner and loser portfolio of the MOM(1,1) strategy. The table shows, quite expectedly,

³⁰These countries are Australia, Belgium, Canada, Denmark, Euro area, France, Germany, Italy, Japan, Netherlands, New Zealand, Norway, Sweden, Switzerland, and the United Kingdom.

that several larger currencies (e.g., Australia, Canada, Japan, New Zealand, Switzerland, United Kingdom) are often included in the momentum strategy but this dominance rests at least partly on the longer sample periods available for these currencies. However, the table also shows large inclusion frequencies for emerging markets such as Brazil, Indonesia, Poland, or Singapore. Hence, it seems worthwhile to investigate whether issues of tradability (or convertibility) affect our results.

As a first exercise, we limit the sample to currencies which have a positive score on the capital account openness index of Chinn and Ito (2006), both in the formation and holding period, to control for the possibility that some currencies are not tradable or that they are only traded in more opaque offshore markets which would not be adequately reflected in the data. We report results for this restricted subset in Table 12, Panel A. As can be seen, the results are not affected by excluding these currencies. Moreover, countries with negative capital account openness index values do not account for a large share of the relevant corner portfolios (less than 20% on average). While a positive score in the Chinn-Ito index already excludes a number of countries (even developed countries, e.g., the U.K. from 1976 to 1978), we additionally run the same exercise under the constraint that a country has to have an index score of at least one. This requirement eliminates several currencies almost completely from the sample (e.g., Brazil, Philippines, Poland, and South Korea) and significantly reduces the investable sample period for other countries (e.g., Belgium only becomes investable in the 1990s). Results for this filter are shown in Panel B of Table 12 but also do not indicate that momentum is primarily driven by currencies which exhibit limitations to investability.

TABLE 12 ABOUT HERE

While the above analysis suggests that tradability issues do not wipe out momentum profits in FX markets, we additionally ran a small survey among four large brokers in FX markets (Goldman Sachs, Deutsche Bank, UBS, Nomura) and asked which currencies would have been impossible (or nearly impossible) to trade in a dynamic portfolio strategy that requires frequent rebalancing. Based on their answers, we restricted our set of tradable currencies

and computed momentum returns on the resulting sample. The following restrictions were imposed: Czech Republic (not tradable before 1999), Hungary (2000), Indonesia (1999), Malaysia (1999), Philipines (1999), Singapore (1999), South Africa (2001), Taiwan (1999), Hong Kong (1986), and Thailand (1999).³¹ Results for this limited sample are shown in Table 13, Panel A. Corroborating the evidence based on the Chinn/Ito index above, we find that momentum profits are still significant after taking into account likely restrictions on tradability of countries.

TABLE 13 ABOUT HERE

As a final check, we augmented the market practitioner’s list by eliminating all currencies with large trading in non-deliverable forwards in offshore markets which may not be adequately covered by our price and interest rate data. These currencies include: Brazil, Egypt, India, Indonesia, South Korea, Malaysia, the Philippines, and Taiwan. Results for this even more restricted set of currencies are shown in Panel B of Table 13 but only strengthen our findings above.

In sum, we find that accounting for capital account restrictions (or other trading restrictions) does not significantly weaken average momentum returns despite excluding many smaller emerging markets from our sample. This finding seems to be driven by the fact that most minor currencies (which are more likely to be subject to capital controls) only enter our sample very recently and, thus, do not drive the lion’s share of our result.

6.2. *Additional tests*

Different base currencies. So far, we have investigated momentum profits from the viewpoint of a U.S. investor. For robustness, we also present results for a British (GBP), Swiss (CHF), Canadian (CAD), and Swedish (SEK) investor, i.e., we convert all data such that

³¹Most of the survey respondents’ other restrictions, for example, regarding Egypt or Saudi Arabia, were actually already reflected in our data where data histories of several currencies start very late at the end of the 1990s or early 2000s (see Table A.1 in the Appendix).

they are quoted against one of these four alternative numeraires. The effective size of the cross-section is, of course, unchanged since we lose one currency (the numeraire) but include the USD as a “new” currency.

Results are shown in Tables A.6 and A.7 in the Appendix for excess returns and spot rate changes, respectively. It can be seen that results are basically unchanged relative to the benchmark case so that momentum is not a U.S. dollar phenomenon. This result is reasonable since our momentum portfolios are dollar-neutral by construction (the USD component cancels out in the long minus short portfolio). Hence, changing the numeraire has little to no effect on the profitability of momentum strategies.

Furthermore, we also run regressions of momentum excess returns for the four different base currencies on a set of risk factors to rule out the possibility that momentum returns are more closely linked to traditional risk factors for non-U.S. investors. Due to data limitations, we cannot obtain data for all risk factors considered in Table 10 so that we focus on the following set of risk factors which should suffice to capture broad economic conditions in these four countries: growth in real industrial production (IP), CPI inflation, growth in real money balances, changes in the term spread, and (local) stock market returns. Results are reported in Table A.8 in the Appendix and we find (similar to the U.S. case in Table 10 above) that momentum returns are not closely linked to any of these standard macro-finance risk factors.

Currency regimes. Another question of relevance is whether momentum strategies can be enhanced by considering information about currency regimes. Intuitively, currencies that are pegged or are only allowed to move in very small bands (or target zones) should be less useful in setting up a momentum strategy than freely floating currencies or currencies that are allowed to move in larger bands. To address this issue we limit our sample of currencies to (i) free floats, managed floats, pre-announced crawling bands (wider than or equal to $\pm 2\%$), de facto crawling bands (narrower than or equal to $\pm 5\%$), moving bands (narrower than or equal to $\pm 2\%$) or (ii) free floats only. Sample (i) corresponds to category 3 whereas sample (ii) corresponds to category 4 of the IMFs (coarse) classification of exchange rate regimes

available on Carmen Reinhart's webpage.³²

Results for these two samples of less heavily managed currencies are shown in Table A.9. The sample period starts in 1986 here to have a large enough cross-section for free floats (also see Figure 1). Panel A reports descriptive statistics for six momentum portfolios and the long minus short portfolio for sample (i). There is a monotonically increasing spread in average excess returns and a significantly positive average excess return for the momentum strategy long in winners and short in losers regardless of the formation period. Panel B shows results for sample (ii) which only comprises free floats. Average excess returns tend to be somewhat lower for formation periods of one and six months but somewhat higher for the 12-months formation period.

In sum, there does not seem to be a clear benefit from concentrating on only freely floating currencies. While freely floating currencies have more room for large price swings, excluding less flexible exchange rates results in a smaller cross-section and excludes a number of slowly trending rates which are managed in crawling bands.

Central bank interventions. Central bank interventions have been considered as one potential source of momentum profits early in the literature. For example, Silber (1994) shows that technical trading rules are more valuable when government agencies intervene in the market. However, later papers reach different conclusions so that the relation between official intervention and momentum trading is less clear-cut. In this vein, Neely (2002) finds that interventions do not influence technical trading profits and that momentum profits are more likely to precede intervention rather than being caused by them.³³

³²<http://www.carmenreinhardt.com/data/browse-by-topic/topics/12>. IMF categories 1 and 2 correspond to more restrictive regimes. It is important to note that for the last several years, the IMF classification of each country (published in the Annual Report on Exchange Arrangements and Exchange Restrictions) is based on the country's actual (de facto) policy, as determined by the IMF. For some countries this classification could differ from the country's official (de jure) stated policy. For most of our sample, only the official stated policy is reported by the IMF.

³³Also, see Neely (1998) for an overview of several findings in the literature on interventions and returns to technical trading. See Sarno and Taylor (2001) for a comprehensive survey on the impact of official intervention on exchange rates.

Given the prominence of this topic in the earlier literature we briefly examine the relationship between intervention and momentum returns in Table A.10 in the Appendix. We report results for regressions of momentum excess returns for our three benchmark strategies on contemporaneous and lagged central bank intervention activity. Intervention activity is proxied for by the sum of absolute intervention amounts of all central banks in the USD (against any foreign currency). Data for this exercise are obtained from the Federal Reserve Bank of St. Louis. Our results show that interventions are not very powerful in capturing momentum returns, broadly consistent with the findings in Neely (2002). However, it should be noted that our analysis is intentionally simple and that there are serious data issues with central bank interventions which are usually not made public.

European Monetary System (EMS). As an additional robustness check, we calculate momentum profits where we exclude all countries participating in the EMS (except for the Deutschmark) and focus on the 1990s where currencies of these countries moved in lock-step.³⁴ Since momentum in any of these countries should be very short-lived it seems likely that excluding these currencies will yield larger momentum profits. We plot cumulative momentum excess returns (for the MOM(1,1) strategy) from 1990 to 1998 in Figure A.2 in the Appendix and do indeed find that excluding EMS member countries leads to a somewhat better performance. Hence, the results reported in the main text seem conservative and it should be possible to increase the profitability of momentum strategies by carefully accounting for the correlation structure of currencies.

7. Conclusion

We have empirically investigated momentum strategies in FX markets, which rely on return continuation among winner and loser currencies. We find that these strategies yield surprisingly high unconditional average excess returns of up to 10% per year and that these returns

³⁴Neely and Weller (1999) investigate returns to technical trading rules in EMS currencies over the period from 1986 to 1996 and find that they generate significant excess returns.

are hard to understand in a framework that relies on covariance risk with standard risk factors. In contrast to an explanation based on systematic risk, we find evidence for under- and subsequent over-reaction in long-horizon momentum returns. In this sense, the evidence for currency momentum seems similar to what has been found for equity markets in earlier literature.

We also find that momentum returns are different from more conventional technical trading rules. As technical trading mostly aims at exploiting trends or momentum in currency movements, it may be expected that returns to these strategies are positively related to our cross-sectional momentum returns. We find, however, that returns to benchmark technical trading rules are somewhat lower and that the correlation with our momentum strategies is rather small. Moreover, currency momentum strategies are very different from the popular carry trade in FX markets. Hence, it comes as no surprise that momentum is not well captured by the global factors that have been shown to be related to carry trade returns in the earlier literature. Rather, momentum and the carry trade are different phenomena which require a different explanation.

However, currency momentum returns do not come as a free lunch for investors trying to exploit these strategies. We find that momentum portfolios in the FX market are significantly skewed towards minor currencies which have relatively high transaction costs, accounting for roughly 50% of momentum returns. Also, the concentration of minor currencies in momentum portfolios raises the need to set up trading positions in currencies with higher idiosyncratic volatility, higher country risk, and higher expected risk of exchange rate instabilities, which clearly imposes risks to investors that are not captured by standard risk factors in a covariance risk framework. Hence, there seem to be effective limits to arbitrage which prevent a straightforward exploitation of momentum returns. Furthermore, momentum profits are highly time-varying, which may also pose an obstacle to arbitrage activity for some of the key FX market participants (e.g. proprietary traders and hedge funds) who typically have fairly short-term investment horizons.

Seen from a broader perspective, there is mounting evidence that momentum can be seen

as an ubiquitous phenomenon in financial markets (e.g. [Asness, Moskowitz, and Pedersen, 2009](#)). A key contribution of this paper is to show that momentum strategies deliver high excess returns in FX markets, comparable in magnitude to the excess returns documented in stock markets. This occurs despite the special characteristics of currency markets, such as huge trading volume ([King and Rime, 2010](#)), mostly professional traders, no short-selling constraints and a considerable degree of central bank interference. However, we show that FX momentum returns are not driven by policy measures including monetary regimes, currency intervention or the implementation of capital account controls. Momentum returns stem primarily from currencies that are hard to hedge and have high country risk, which is similar to recent findings that equity momentum is concentrated in stocks with high credit risk ([Avramov, Chordia, Jostova, and Philipov, 2007](#)), and momentum in corporate bonds is concentrated in non-investment grade bonds ([Jostova, Nikolova, Philipov, and Stahel, 2010](#)). In sum, these findings suggest that there may be a common source of momentum profits across asset classes.

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Table 1. Momentum returns and Sharpe Ratios

This table shows annualized average returns for different momentum strategies ($\overline{r^{f,h}}$) in Panel A. The rows show formation periods (f) whereas the columns indicate holding periods (h) in months. Numbers in brackets are t-statistics based Newey-West HAC standard errors. The left part of the table shows currency excess returns (spot rate changes adjusted for interest rate differentials) whereas the right part shows pure spot rate returns. Panel B shows annualized Sharpe Ratios. T-statistics based on a moving block-bootstrap are in squared brackets. The right panel shows average annualized spot rate changes (in percent) divided by the annualized standard deviation of mean exchange rate changes. The sample period is January 1976 – January 2010 and we employ monthly returns.

Panel A. Excess returns and spot rate changes											
Excess returns						Spot rate changes					
Holding period h						Holding period h					
f	1	3	6	9	12	f	1	3	6	9	12
1	9.46 [5.31]	7.00 [4.11]	6.17 [3.13]	5.15 [2.73]	5.75 [3.6]	1	7.91 [4.55]	4.42 [3.07]	3.38 [1.93]	4.75 [2.94]	3.13 [2.02]
3	9.40 [5.30]	6.32 [3.80]	4.96 [3.03]	4.67 [2.92]	4.43 [2.74]	3	8.54 [5.10]	5.73 [3.59]	5.28 [3.66]	4.63 [2.88]	5.10 [3.51]
6	8.54 [4.78]	6.31 [3.63]	3.66 [2.06]	3.25 [1.79]	3.14 [1.69]	6	6.50 [3.88]	5.75 [4.00]	3.47 [2.15]	3.64 [2.32]	3.17 [1.80]
9	7.18 [3.80]	6.80 [3.65]	5.36 [2.86]	3.86 [2.05]	3.24 [1.67]	9	8.33 [4.82]	7.06 [4.23]	6.50 [3.91]	4.91 [2.87]	4.09 [2.35]
12	6.16 [3.40]	5.48 [3.24]	3.02 [1.75]	2.05 [1.17]	1.89 [1.04]	12	7.59 [4.63]	6.04 [4.02]	3.94 [2.59]	3.19 [1.97]	3.03 [1.92]

Panel B. Sharpe Ratios and normalized spot rate changes											
Excess returns						Spot rate changes					
Holding period h						Holding period h					
f	1	3	6	9	12	f	1	3	6	9	12
1	0.95 [5.48]	0.76 [4.10]	0.59 [3.15]	0.56 [2.47]	0.61 [2.95]	1	0.84 [5.52]	0.53 [4.23]	0.37 [3.25]	0.57 [2.81]	0.37 [3.21]
3	0.88 [5.37]	0.60 [3.70]	0.50 [3.04]	0.53 [2.74]	0.51 [2.42]	3	0.86 [5.17]	0.57 [3.73]	0.58 [3.45]	0.50 [2.99]	0.63 [2.61]
6	0.79 [4.55]	0.60 [3.53]	0.37 [1.94]	0.34 [1.76]	0.33 [1.48]	6	0.64 [4.76]	0.60 [3.70]	0.38 [2.06]	0.41 [2.05]	0.35 [1.43]
9	0.67 [3.76]	0.63 [3.61]	0.50 [2.95]	0.36 [1.95]	0.30 [1.57]	9	0.85 [3.99]	0.71 [3.66]	0.66 [3.07]	0.51 [2.12]	0.41 [1.84]
12	0.61 [3.18]	0.56 [3.05]	0.32 [1.64]	0.21 [1.17]	0.19 [1.05]	12	0.77 [3.48]	0.64 [3.32]	0.44 [1.89]	0.35 [1.27]	0.33 [1.14]

Table 2. Momentum²

This table shows average momentum excess returns and Sharpe Ratios (SR) for momentum strategies based on other momentum portfolios. We first calculate monthly excess returns for all 144 possible momentum portfolios based on formation and holding periods of $f = 1, 2, \dots, 12$ and $h = 1, 2, \dots, 12$. Next we run a momentum strategy on these 144 momentum portfolios and sort momentum strategies from the first step into nine portfolios based on their lagged returns over an evaluation period. Lagged returns over the evaluation period (shown in the first column) vary from one to 120 months. We report results for the nine portfolios from the second stage (from “Worst” lagged strategy to “Best” lagged strategy) and a high minus low portfolio (best strategy minus worst strategy, “B-W”) that is long in the best 16 (144/9) strategies and short in the worst 16 (144/9) strategies from the first step. Numbers in brackets are t -statistics based on [Newey and West \(1987\)](#).

		Momentum ² –Portfolios									
Lag		Worst	2	3	4	5	6	7	8	Best	B-W
1	Mean	1.03	2.45	3.44	3.78	4.34	5.46	5.82	6.85	8.70	7.67
	t	[0.79]	[1.74]	[2.40]	[2.54]	[2.90]	[3.50]	[3.76]	[4.46]	[5.68]	[4.31]
	SR	0.14	0.31	0.42	0.45	0.49	0.62	0.63	0.75	0.94	0.73
3	Mean	1.21	3.02	3.85	3.53	4.51	5.14	5.72	6.67	7.62	6.41
	t	[1.04]	[2.28]	[2.71]	[2.46]	[3.03]	[3.37]	[3.72]	[4.24]	[5.05]	[4.12]
	SR	0.20	0.43	0.50	0.43	0.53	0.58	0.62	0.72	0.83	0.71
6	Mean	3.40	2.99	3.47	3.92	3.93	4.69	6.09	6.54	7.88	4.48
	t	[2.87]	[2.39]	[2.63]	[2.87]	[2.62]	[3.17]	[3.94]	[4.27]	[5.04]	[3.08]
	SR	0.54	0.45	0.48	0.52	0.47	0.54	0.67	0.72	0.85	0.54
9	Mean	3.59	3.82	4.04	4.61	4.81	5.28	5.55	6.07	7.10	3.51
	t	[3.02]	[3.11]	[3.13]	[3.43]	[3.44]	[3.43]	[3.69]	[3.89]	[4.52]	[2.53]
	SR	0.57	0.57	0.56	0.60	0.59	0.60	0.63	0.68	0.79	0.44
12	Mean	3.29	4.57	4.57	4.42	4.86	4.98	5.54	5.87	7.25	3.96
	t	[2.76]	[3.69]	[3.39]	[3.28]	[3.46]	[3.51]	[3.44]	[3.75]	[4.83]	[3.14]
	SR	0.53	0.66	0.62	0.58	0.59	0.59	0.61	0.66	0.81	0.51
60	Mean	3.83	3.75	4.24	4.82	4.15	5.28	5.18	5.42	5.92	2.09
	t	[2.70]	[2.60]	[2.90]	[3.27]	[2.75]	[3.62]	[3.25]	[3.49]	[3.86]	[1.88]
	SR	0.55	0.53	0.58	0.63	0.52	0.66	0.60	0.64	0.68	0.33
120	Mean	3.69	4.21	4.82	4.60	4.54	5.01	5.75	5.76	6.40	2.70
	t	[2.16]	[2.48]	[2.80]	[2.61]	[2.62]	[2.88]	[3.20]	[3.18]	[3.59]	[2.03]
	SR	0.49	0.56	0.61	0.55	0.55	0.61	0.68	0.65	0.74	0.41

Table 3. Moving average rules and cross-sectional momentum

This table shows means, Sharpe Ratios (SR), standard deviations, skewness, and kurtosis for excess returns to three benchmark moving average (MA) rules in Panel A. Panel B shows results from regressions of cross-sectional momentum excess returns (i.e., high-minus-low portfolios) on a constant and excess returns to each of the three MA rules. Note that the adjusted R^2 s in Panel B are in percent.

Panel A. Descriptive statistics for MA rules									
	(1, 20)	(5, 20)	(1, 200)						
Mean	5.27	5.14	5.23						
	[5.56]	[5.73]	[4.64]						
SR	0.88	0.83	0.77						
St. Dev.	5.98	6.22	6.81						
Skewness	0.67	0.40	0.09						
Kurtosis	4.63	4.71	4.97						
Panel B. Regressions of cross-sectional momentum returns on MA rule returns									
	MOM(1,1)			MOM(6,1)			MOM(12,1)		
$\alpha_{(1,20)}$	7.74			7.63			6.21		
	[4.54]			[4.60]			[3.62]		
$\beta_{(1,20)}$	0.33			0.17			-0.01		
	[3.57]			[1.41]			[-0.12]		
$\alpha_{(5,20)}$		7.80			7.49			5.84	
		[4.52]			[4.45]			[3.35]	
$\beta_{(5,20)}$		0.32			0.21			0.06	
		[3.95]			[1.72]			[0.60]	
$\alpha_{(1,200)}$			6.90		4.16				7.39
			[3.97]		[2.46]				[2.82]
$\beta_{(1,200)}$			0.47		0.81				0.03
			[5.67]		[7.56]				[0.16]
$\bar{R}^2$ (in %)	3.62	3.88	10.34	0.68	1.17	26.00	-0.25	-0.10	-0.23

Table 4. Comparing momentum and carry trade portfolios

This Table shows descriptive statistics for six momentum (Panel A) and six carry trade portfolios (Panel B). Currencies are sorted into six portfolios depending on their lagged one month excess return rx_{-1} (momentum portfolios) or their lagged forward discount $(f - s)_{-1}$ (carry trade portfolios). The 1/6 (16.67%) of all currencies with the lowest lagged excess return (or forward discount) are allocated to portfolio “Low”, whereas the 1/6 of all currencies with the highest lagged excess returns (or forward discounts) are allocated to portfolio “High”. Portfolios 2–5 each consist of 1/6 of all currencies and have increasingly higher lagged excess returns (or forward discounts). Portfolios are rebalanced monthly. We also report results for an the average of all six portfolios (“Av.”) and a portfolio that is long in portfolio “High” and short in portfolio “Low” (“H–L”). Shown are average annualized excess returns, the standard deviation, skewness, and kurtosis of excess returns. The last two rows of each panel show average lagged excess returns $\overline{rx}_{-1}$ and forward discounts $\overline{(f - s)}_{-1}$ for currencies in each portfolio at the time of portfolio formation. Also shown are average returns across the six portfolios (“Av.”) and the difference between the “High” and “Low” portfolios (“H–L”). The sample period is Januar 1976 – January 2010.

Panel A: Momentum Portfolios ($f = 1, h = 1$)								
	Low	2	3	4	5	High	Av.	H–L
Mean	-4.17 [-2.36]	-0.87 [-0.49]	0.27 [0.16]	2.25 [1.31]	2.08 [1.25]	5.28 [2.94]	0.81 [0.53]	9.46 [5.26]
Stand. Dev.	2.88	2.57	2.61	2.57	2.64	2.64	2.28	2.87
Skewness	-0.27	-0.79	-0.32	-0.26	-0.58	-0.29	-0.42	0.06
Kurtosis	5.97	6.38	4.45	4.61	6.78	4.49	4.48	5.29
$\overline{rx}_{-1}$	-2.93	-1.03	-0.23	0.42	1.21	2.94		
$\overline{(f - s)}_{-1}$	0.44	0.75	1.17	1.34	1.93	5.13		
Panel B: Carry Trade Portfolios								
	Low	2	3	4	5	High	Av.	H–L
Mean	-3.39 [-1.94]	-1.41 [-0.93]	0.24 [0.15]	1.32 [0.81]	2.04 [1.17]	6.77 [3.22]	0.93 [0.61]	10.15 [5.79]
Stand. Dev.	2.71	2.39	2.39	2.49	2.64	2.98	2.28	2.64
Skewness	-0.21	-0.42	-0.28	-0.37	-0.75	-0.35	-0.37	-0.69
Kurtosis	4.85	4.34	5.58	5.12	5.84	4.33	4.34	4.20
$\overline{rx}_{-1}$	-0.32	-0.11	0.01	0.13	0.23	0.52		
$\overline{(f - s)}_{-1}$	-4.81	-1.79	0.02	1.59	4.02	11.65		

Table 5. Correlation of momentum and carry trade returns

This Table shows correlation coefficients between portfolio returns. Panel A shows correlation coefficients between momentum returns based on strategies with formation horizons of f equal to one, six, and twelve months and holding periods of $h = 1$ month (denoted $MOM_{1,1}$, $MOM_{6,1}$, $MOM_{12,1}$, respectively) and forward discount-sorted portfolio returns (denoted C since they form the basis of the carry trade). Returns are based on six portfolios and a long-short portfolio for both momentum and the carry trade. We only report correlations for corresponding pairs of portfolios. For example, in row $\rho(M_{1,1}, C)$ we report the correlation of the “Low” momentum portfolio with the “Low” carry trade portfolio in column “Low”, the correlation of the third momentum portfolio with the third carry trade portfolio, and so on for all six portfolios and the long-short portfolios. Row $\rho(M_6, C)$ shows the correlations between portfolios pairs of the momentum strategy with a six months formation period with the carry trade and row $\rho(M_{12}, C)$ shows the correlations between portfolio pairs of the twelve months formation period momentum strategy and the carry trade. Panel B shows correlations for momentum portfolios with different formation horizons.

Panel A: Momentum and carry trade portfolios							
	Low	2	3	4	5	High	H-L
$\rho(MOM_{1,1}, C)$	0.68	0.84	0.83	0.85	0.81	0.73	0.04
$\rho(MOM_{6,1}, C)$	0.63	0.84	0.82	0.83	0.81	0.74	0.01
$\rho(MOM_{12,1}, C)$	0.67	0.85	0.81	0.87	0.82	0.74	0.07
Panel B: Momentum portfolios							
	Low	2	3	4	5	High	H-L
$\rho(MOM_{1,1}, MOM_{6,1})$	0.77	0.83	0.88	0.85	0.83	0.79	0.45
$\rho(MOM_{1,1}, MOM_{12,1})$	0.66	0.81	0.86	0.87	0.80	0.78	0.28
$\rho(MOM_{6,1}, MOM_{12,1})$	0.82	0.89	0.89	0.89	0.91	0.89	0.73

Table 6. Double sorts

This Table shows annualized mean excess returns for double-sorted portfolios. All currencies in the sample are first sorted on lagged forward discounts (FD) into two portfolios along the median. Next, currencies within each of the two subgroups are allocated into three momentum portfolios depending on their lagged excess returns over $f = 1, 6$, or 12 months. Hence, row FD_L denotes the 50% of all currencies with the lowest (lagged) forward discount whereas FD_H denotes the 50% of all currencies with the highest (lagged) forward discounts. Columns M_L , M_M , and M_H denote the 33% of all currencies with the lowest, intermediate, and the highest (lagged) returns, respectively. Columns Δ_M shows the return difference between high and low momentum portfolios ($M_H - M_L$) for each subgroup of currencies whereas e.g. Δ_{FD} shows the return difference between the forward discount-sorted portfolios for each momentum subgroup. The lower-right cell in each sub-panel shows the return difference between the two momentum “high minus low” portfolios of each forward discount category. We report annualized excess returns in percent for each portfolio and all high-minus-low portfolios. Numbers in brackets are HAC t-statistics and the sample runs from January 1976 – January 2010.

Carry Trade and Momentum												
$f = 1, h = 1$					$f = 6, h = 1$					$f = 12, h = 1$		
	M_L	M_M	M_H	Δ_M	M_L	M_M	M_H	Δ_M	M_L	M_M	M_H	Δ_M
FD_L	-4.52 [-2.90]	-0.90 [-0.55]	0.54 [0.34]	5.06 [3.81]	-4.40 [-2.81]	-0.35 [-0.21]	0.06 [0.04]	4.46 [3.63]	-3.94 [-2.34]	-0.40 [-0.24]	0.09 [0.06]	4.04 [2.86]
FD_H	0.64 [0.34]	3.20 [1.68]	6.00 [3.18]	5.36 [3.30]	2.38 [1.14]	2.43 [1.45]	6.34 [3.29]	3.96 [2.43]	2.86 [1.49]	3.21 [1.80]	5.98 [3.10]	3.12 [2.02]
Δ_{FD}	5.16 [4.00]	4.10 [3.43]	5.45 [3.89]	0.30 [0.17]	6.77 [4.33]	2.78 [2.57]	6.27 [4.58]	-0.50 [-0.26]	6.80 [4.71]	3.61 [3.22]	5.89 [4.56]	-0.91 [-0.49]

Table 7. Cross-sectional regressions

This Table shows results for cross-sectional regressions of individual currencies' excess returns (left part) or spot rate changes (right part) on lagged excess returns, lagged forward discounts, and/or lagged spot rate changes. Numbers in parentheses are standard errors of the cross-sectional R^2 s. For ease of interpretation we have multiplied spot rate changes by minus one so that higher values indicate an appreciation of the foreign currency against the USD.

Panel A: One month									
Dependent: Excess returns					Dependent: Spot rate changes				
const.	rx	$f - s$	Δs	R^2	const.	rx	$f - s$	Δs	R^2
-0.02	0.16			0.15	-0.16	0.08			0.13
[-0.17]	[5.65]			(0.01)	[-1.52]	[2.95]			(0.01)
0.00		0.63		0.14	0.00		-0.37		0.09
[0.01]		[4.87]		(0.01)	[0.01]		[-2.89]		(0.01)
0.02			0.13	0.13	-0.16			0.13	0.14
[0.22]			[4.46]	(0.01)	[-1.59]			[4.55]	(0.01)
-0.07	0.12	0.57		0.26	-0.07	0.12	-0.43		0.20
[-0.76]	[4.42]	[4.68]		(0.01)	[-0.76]	[4.42]	[-3.52]		(0.01)
-0.07		0.68	0.14	0.26	-0.07		-0.32	0.14	0.21
[-0.72]		[5.89]	[4.82]	(0.01)	[-0.72]		[-2.83]	[4.82]	(0.01)
Panel B: Six months									
0.06	0.30			0.17	-0.05	0.15			0.15
[0.57]	[5.65]			(0.01)	[-0.46]	[3.07]			(0.01)
0.04		0.46		0.13	0.04		-0.52		0.09
[0.33]		[2.98]		(0.01)	[0.31]		[-3.33]		(0.01)
0.12			0.19	0.14	-0.03			0.25	0.15
[1.20]			[3.24]	(0.01)	[-0.30]			[4.87]	(0.01)
0.08	0.21	0.36		0.27	0.07	0.23	-0.64		0.24
[0.82]	[3.89]	[2.36]		(0.02)	[0.82]	[4.39]	[-4.20]		(0.01)
0.06		0.57	0.23	0.27	0.07		-0.41	0.23	0.24
[0.71]		[4.01]	[4.27]	(0.02)	[0.77]		[-2.90]	[4.33]	(0.01)
Panel C: Twelve months									
-0.05	0.28			0.16	-0.17	0.12			0.15
[-0.52]	[3.97]			(0.01)	[-1.66]	[1.79]			(0.01)
0.04		0.42		0.12	0.03		-0.51		0.09
[0.36]		[2.66]		(0.01)	[0.29]		[-3.22]		(0.01)
0.03			0.20	0.14	-0.05			0.32	0.14
[0.24]			[2.45]	(0.01)	[-0.47]			[4.52]	(0.01)
-0.06	0.20	0.28		0.25	-0.06	0.25	-0.66		0.24
[-0.66]	[2.58]	[1.74]		(0.01)	[-0.62]	[3.21]	[-4.06]		(0.01)
-0.04		0.48	0.24	0.25	-0.04		-0.42	0.27	0.24
[-0.47]		[3.21]	[3.14]	(0.01)	[-0.42]		[-2.70]	[3.41]	(0.01)

Table 8. Momentum returns after transaction costs

This table shows annualized average returns for different momentum strategies $(\overline{r^{f,h}})$ after adjusting for bid-ask spreads. The rows show formation periods (f) whereas the columns indicate holding periods (h). The formation and holding period can be 1, 3, 6, 9, or 12 months, respectively. Numbers in brackets are t-statistics based Newey-West standard errors. The left part of the table shows net currency excess returns (spot rate changes adjusted for interest rate differentials) whereas the right part shows net spot rate returns. The sample period is January 1976 – January 2010 and we employ monthly returns.

Net excess returns						Net spot rate changes					
f	Holding period h					f	Holding period h				
	1	3	6	9	12		1	3	6	9	12
1	3.92	2.02	1.26	0.38	0.39	1	4.84	3.36	2.69	4.43	2.53
	[2.20]	[1.16]	[0.61]	[0.18]	[0.20]		[2.81]	[2.37]	[1.57]	[2.76]	[1.65]
3	4.41	2.12	0.88	0.97	-0.07	3	6.80	4.58	4.72	4.33	4.86
	[2.39]	[1.20]	[0.53]	[0.58]	[-0.04]		[3.99]	[2.81]	[3.18]	[2.58]	[3.32]
6	3.86	2.12	-0.27	-0.92	-1.28	6	5.06	4.83	3.06	3.27	3.29
	[2.09]	[1.19]	[-0.15]	[-0.49]	[-0.67]		[3.03]	[3.37]	[1.94]	[2.08]	[1.88]
9	2.48	2.43	0.99	-0.40	-1.06	9	7.53	6.73	6.19	4.81	3.84
	[1.26]	[1.27]	[0.51]	[-0.21]	[-0.54]		[4.34]	[4.00]	[3.69]	[2.88]	[2.20]
12	1.40	0.80	-1.46	-1.98	-2.44	12	6.65	5.53	3.75	2.92	2.77
	[0.74]	[0.45]	[-0.84]	[-1.11]	[-1.31]		[4.01]	[3.66]	[2.47]	[1.79]	[1.73]

Table 9. Momentum returns with effective spreads of 75% and 50%

This table reports transaction cost adjusted excess returns with effective spreads of 75% (Panel A) and 50% (Panel B) of the quoted spread, respectively. The table setup is the same as in Table 1 but we only show results for excess returns (and not for spot rate changes). The sample period is January 1976 – January 2010 and we employ monthly returns.

Panel A: Effective spread of 75%						Panel B: Effective spread of 50%					
f	Holding period h					f	Holding period h				
	1	3	6	9	12		1	3	6	9	12
1	5.28	3.24	2.51	1.53	1.69	1	6.64	4.47	3.77	2.69	3.00
	[2.98]	[1.89]	[1.25]	[0.76]	[0.88]		[3.76]	[2.62]	[1.89]	[1.36]	[1.61]
3	5.61	3.16	1.86	1.85	0.97	3	6.81	4.20	2.83	2.74	2.00
	[3.07]	[1.82]	[1.12]	[1.12]	[0.59]		[3.76]	[2.45]	[1.72]	[1.68]	[1.23]
6	5.03	3.17	0.70	0.15	-0.18	6	6.20	4.23	1.68	1.21	0.92
	[2.76]	[1.80]	[0.39]	[0.08]	[-0.10]		[3.43]	[2.41]	[0.94]	[0.66]	[0.49]
9	3.66	3.56	2.16	0.68	0.08	9	4.85	4.69	3.33	1.75	1.24
	[1.89]	[1.89]	[1.13]	[0.35]	[0.04]		[2.53]	[2.52]	[1.76]	[0.93]	[0.64]
12	2.60	1.97	-0.35	-0.94	-1.36	12	3.80	3.13	0.78	0.09	-0.28
	[1.39]	[1.12]	[-0.20]	[-0.53]	[-0.74]		[2.07]	[1.81]	[0.45]	[0.05]	[-0.15]

Table 10. Macro risk

This Table shows time-series regression estimates of currency momentum returns (long-short portfolios $MOM_{1,1}$, $MOM_{6,1}$, and $MOM_{12,1}$) on various macro factors and other risk factors. Consumption is real consumption growth, Employment denotes U.S. total nonfarm employment growth, ISM denotes the ISM manufacturing index, IP denotes growth in real industrial production, CPI denotes the inflation rate, M2 is the growth in real money balances, Disp Inc is growth in real disposable personal income, TED denotes the TED spread, Term denotes the term spread (20 years minus 3 months), HML_{FX} is the return to the carry trade long-short portfolio (Lustig, Roussanov, and Verdelhan, 2011), and VOL_{FX} is a proxy for global FX volatility (Menkhoff, Sarno, Schmeling, and Schrimpf, 2011). MKTRF, HML, and SMB are the Fama-French factors and UMD denotes the return to a long-short U.S. momentum portfolio. Panel A shows results for univariate regressions (intercepts α , slope coefficients β , and the adjusted R^2) whereas the Panel B shows results from a multivariate regression of momentum returns on the three Fama-French factors and UMD. Bold numbers indicate significance at the 5%-level or below.

Panel A: Univariate regressions									
	$MOM_{1,1}$			$MOM_{6,1}$			$MOM_{12,1}$		
	α	β	R^2	α	β	R^2	α	β	R^2
Consumption	9.65	-0.05	0.00	8.95	-0.12	0.00	6.03	0.07	0.00
Employment	10.57	-0.72	0.00	7.74	0.62	0.00	5.86	0.23	0.00
ISM	9.46	0.04	0.00	8.60	0.03	0.00	6.14	0.04	0.00
IP	9.72	0.11	0.00	8.72	0.04	0.00	6.26	0.03	0.00
CPI	11.73	-0.55	0.00	9.11	-0.12	0.00	6.60	-0.10	0.00
M2	9.97	0.34	0.00	8.68	0.02	0.00	6.18	-0.01	0.00
Disp Inc	9.33	0.07	0.00	8.42	0.10	0.00	5.95	0.10	0.00
TED	13.64	-0.38	0.01	11.95	-0.30	0.01	9.73	-0.32	0.01
Term	4.48	0.22	0.01	7.54	0.05	0.00	5.05	0.05	0.00
HML_{FX}	9.50	0.04	0.00	8.65	0.02	0.00	6.21	0.08	0.00
VOL_{FX}	11.70	-0.44	0.00	18.75	-2.04	0.01	27.59	-4.29	0.04
Panel B: Multivariate regressions									
	$MOM_{1,1}$			$MOM_{6,1}$			$MOM_{12,1}$		
	α	β	R^2	α	β	R^2	α	β	R^2
MKTRF	8.73	0.00	0.00	8.02	0.04	0.00	5.16	0.02	0.00
SMB		0.97			-0.54			0.71	
HML		0.06			0.01			0.06	
UMD		0.02			0.03			0.04	

Table 11. Double sorts on idiosyncratic volatility or risk ratings and momentum

The setup of this Table is identical to Table 6 but here we sort on idiosyncratic volatility and momentum (Panel A), country risk and momentum (Panel B), and exchange rate stability risk and momentum (Panel C).

Panel A: Idiosyncratic Volatility and Momentum												
$f = 1, h = 1$					$f = 6, h = 1$					$f = 12, h = 1$		
	M_L	M_M	M_H	Δ_M	M_L	M_M	M_H	Δ_M	M_L	M_M	M_H	Δ_M
$IVOL_L$	-1.04 [-0.65]	0.92 [0.55]	2.93 [1.75]	3.97 [2.81]	-0.85 [-0.50]	1.08 [0.66]	2.82 [1.79]	3.67 [3.04]	0.15 [0.10]	1.13 [0.67]	2.27 [1.31]	2.12 [1.58]
$IVOL_H$	-3.52 [-1.83]	1.00 [0.57]	4.57 [2.48]	8.09 [4.72]	-2.22 [-1.16]	0.24 [0.14]	4.77 [2.44]	6.99 [4.28]	-0.78 [-0.41]	0.20 [0.11]	4.38 [2.30]	5.16 [3.01]
Δ_{IVOL}	-2.48 [-1.86]	0.07 [0.07]	1.64 [1.28]	4.11 [2.18]	-1.38 [-1.15]	-0.84 [-0.86]	1.95 [1.52]	3.33 [2.05]	-0.93 [-0.80]	-0.94 [-0.89]	2.11 [1.63]	3.04 [1.79]

Panel B: Country Risk and Momentum

	M_L	M_M	M_H	Δ_M	M_L	M_M	M_H	Δ_M	M_L	M_M	M_H	Δ_M
$CRISK_L$	0.01 [0.01]	3.41 [1.78]	4.51 [2.52]	4.50 [3.12]	0.95 [0.49]	3.14 [1.67]	4.26 [2.33]	3.31 [2.67]	1.65 [0.80]	3.24 [1.67]	3.86 [2.10]	2.21 [1.51]
$CRISK_H$	-0.67 [-0.34]	3.82 [1.90]	8.04 [3.72]	8.72 [4.19]	0.89 [0.40]	3.39 [1.94]	7.24 [3.24]	6.35 [2.92]	2.58 [1.29]	2.70 [1.43]	8.65 [3.56]	6.07 [2.34]
Δ_{CRISK}	-0.68 [-0.46]	0.41 [0.35]	3.53 [2.21]	4.22 [2.12]	-0.06 [-0.04]	0.25 [0.20]	2.97 [2.02]	3.04 [1.72]	0.93 [0.54]	-0.54 [-0.45]	3.79 [2.61]	3.87 [1.93]

Panel C: Exchange Rate Stability Risk and Momentum

	M_L	M_M	M_H	Δ_M	M_L	M_M	M_H	Δ_M	M_L	M_M	M_H	Δ_M
$XSTAB_L$	1.27 [0.83]	0.15 [0.10]	3.25 [2.17]	1.98 [1.39]	1.56 [0.96]	0.30 [0.23]	3.60 [2.32]	2.04 [1.31]	0.80 [0.50]	1.40 [1.04]	3.22 [2.15]	2.42 [1.70]
$XSTAB_H$	-0.48 [-0.24]	4.04 [2.02]	6.09 [3.09]	6.56 [4.06]	0.51 [0.24]	3.35 [1.77]	6.06 [2.93]	5.55 [3.31]	1.58 [0.77]	3.38 [1.80]	6.36 [3.12]	4.78 [2.50]
Δ_{XSTAB}	-1.75 [-1.06]	3.89 [2.47]	2.84 [1.58]	4.59 [2.44]	-1.05 [-0.59]	3.05 [2.01]	2.47 [1.31]	3.51 [1.70]	0.78 [0.43]	1.98 [1.21]	3.14 [1.82]	2.35 [1.11]

Table 12. Momentum returns and capital controls

The setup of this table is identical to Table 1 but here we exclude countries with capital controls. More specifically, at each point in time, we only include currencies of countries which have a score in excess of zero (Panel A) or a score higher than one (Panel B) in the capital account openness index of Chinn and Ito (which is based on an update of the data in Chinn and Ito (2006)).

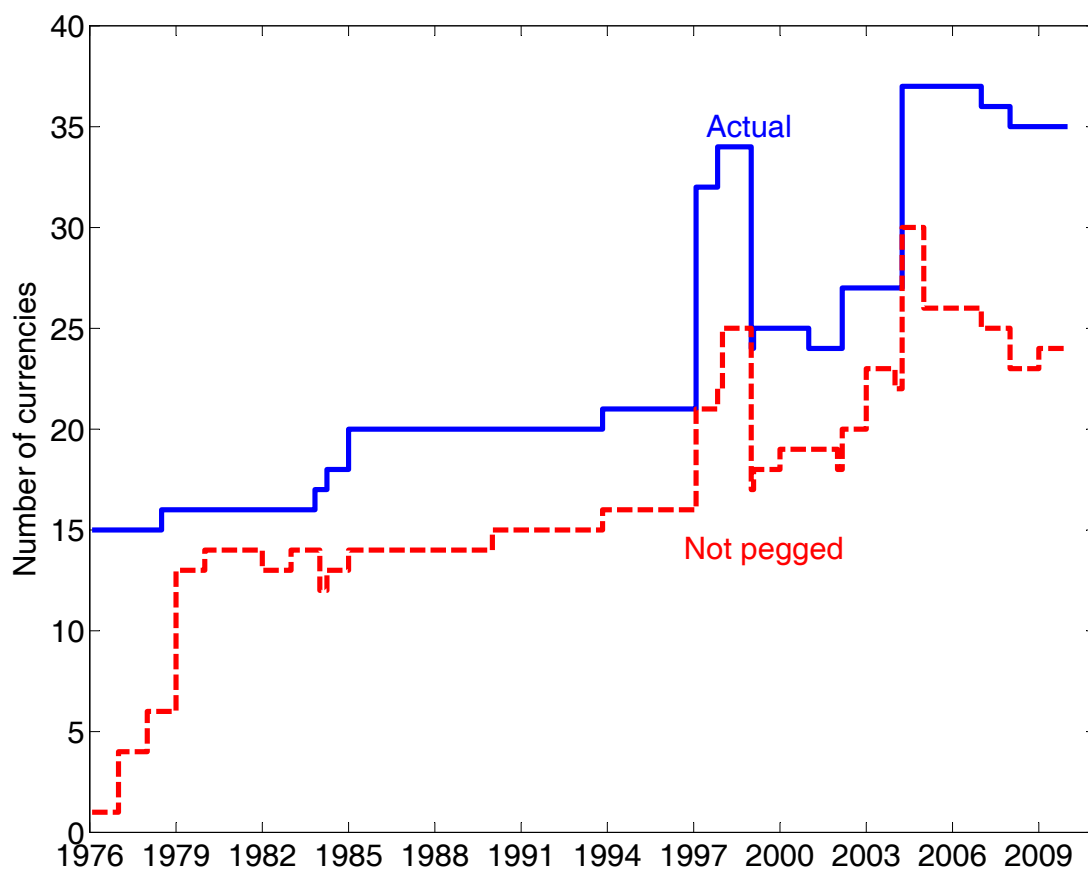
Excess returns (without b/a)						Spot rate changes (without b/a)					
Holding period h						Holding period h					
f	1	3	6	9	12	f	1	3	6	9	12
Panel A. Chinn-Ito Index > 0											
1	9.22	6.46	6.02	4.54	4.63	1	5.14	3.07	3.75	2.84	2.90
	[4.61]	[3.47]	[2.87]	[2.42]	[2.39]		[2.79]	[1.86]	[1.86]	[1.58]	[1.57]
3	9.90	7.07	6.18	5.39	5.33	3	7.04	5.33	4.91	3.62	4.00
	[5.30]	[3.96]	[3.42]	[3.44]	[3.07]		[3.81]	[2.81]	[2.70]	[2.06]	[2.30]
6	9.65	8.11	4.71	3.42	4.22	6	6.48	7.29	3.32	2.14	1.33
	[5.29]	[4.52]	[2.52]	[1.74]	[2.26]		[3.30]	[4.11]	[1.72]	[1.14]	[0.68]
9	8.95	8.46	5.60	4.53	2.64	9	6.83	5.76	5.30	3.64	1.82
	[4.53]	[4.48]	[2.85]	[2.21]	[1.37]		[3.42]	[2.81]	[2.66]	[1.79]	[0.92]
12	6.90	6.51	3.77	2.54	1.40	12	4.61	4.20	1.73	1.42	-0.53
	[3.61]	[3.58]	[2.06]	[1.38]	[0.77]		[2.33]	[2.20]	[0.96]	[0.78]	[-0.28]
Panel B. Chinn-Ito Index > 1											
1	8.97	5.80	6.51	4.65	4.25	1	5.06	2.96	4.10	2.27	2.75
	[4.61]	[3.01]	[3.07]	[2.48]	[2.02]		[2.79]	[1.77]	[2.02]	[1.25]	[1.45]
3	9.85	7.12	5.98	5.53	5.34	3	7.51	5.41	4.74	4.43	3.52
	[5.10]	[3.95]	[3.28]	[3.54]	[2.87]		[3.91]	[2.85]	[2.70]	[2.51]	[2.08]
6	10.50	8.86	5.40	3.43	2.34	6	6.19	6.61	3.17	2.43	-0.39
	[5.45]	[4.76]	[2.83]	[1.55]	[1.32]		[3.02]	[3.46]	[1.58]	[1.24]	[-0.20]
9	9.22	8.16	5.51	4.87	2.46	9	6.69	5.46	5.19	2.83	1.80
	[4.47]	[3.97]	[2.66]	[2.11]	[1.18]		[3.18]	[2.55]	[2.48]	[1.35]	[0.91]
12	6.88	6.05	2.63	2.18	1.54	12	4.73	3.72	1.46	1.09	-0.20
	[3.42]	[3.09]	[1.16]	[1.21]	[0.83]		[2.34]	[1.86]	[0.77]	[0.61]	[-0.10]

Table 13. Momentum and tradability

This table shows average annualized excess returns for six momentum portfolios sorted on lagged one, six, and twelve month returns and the corresponding high minus low momentum portfolios (H-L). Panel A shows results for a set of investable currencies as identified in a survey of FX professionals in major investment banks. Panel B additionally excludes all currencies with non-deliverable forward trading in offshore markets. We refer to the main text for details. Numbers in brackets are t -statistics based on [Newey and West \(1987\)](#) and the sample period is 1976 to 2010.

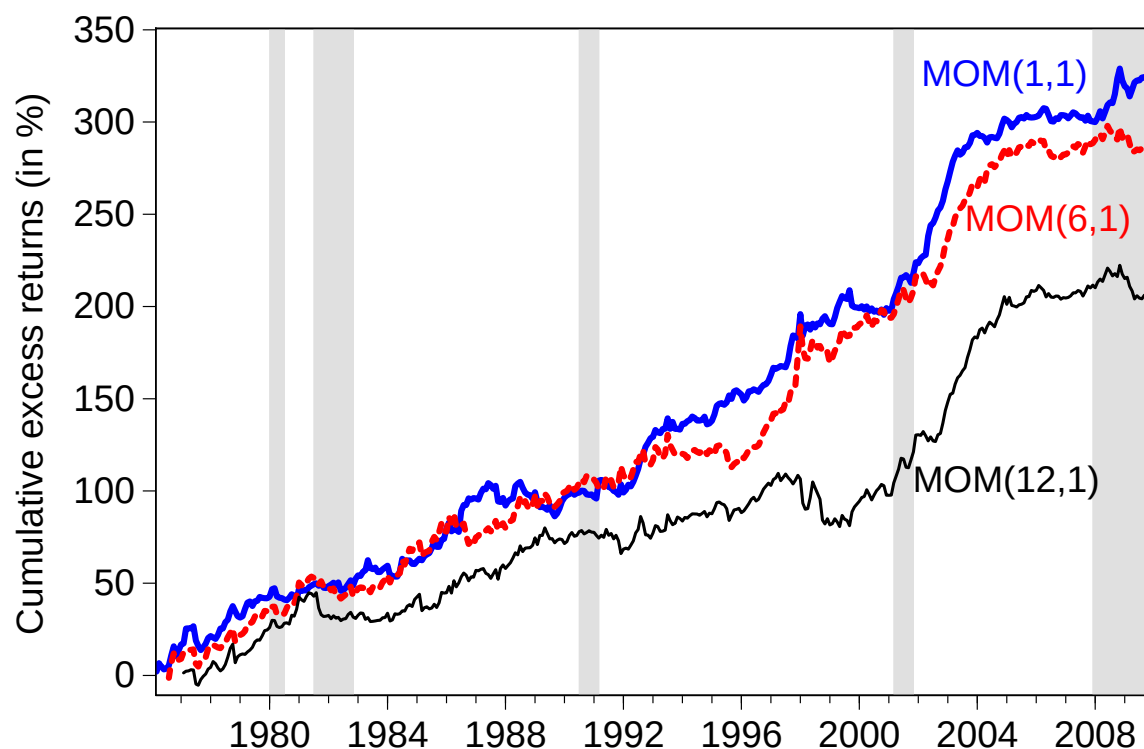
Panel A. "Investable" currency universe							
f	L	2	3	4	5	H	H-L
1	-3.28 [-1.89]	-0.17 [-0.09]	0.67 [0.38]	2.58 [1.45]	2.47 [1.50]	5.48 [2.99]	8.76 [4.90]
6	-1.85 [-1.05]	-0.79 [-0.44]	1.45 [0.88]	1.16 [0.68]	2.31 [1.36]	5.81 [2.98]	7.65 [4.80]
12	-2.14 [-1.15]	0.38 [0.22]	0.94 [0.55]	1.43 [0.83]	2.97 [1.69]	4.75 [2.55]	6.89 [4.05]
Panel B. "Investable" currency universe ex NDF							
f	L	2	3	4	5	H	H-L
1	-2.80 [-1.64]	0.31 [0.17]	0.98 [0.52]	1.99 [1.07]	2.58 [1.56]	5.30 [2.88]	8.10 [4.69]
6	-1.68 [-0.97]	-0.20 [-0.11]	1.53 [0.87]	1.15 [0.65]	2.38 [1.38]	5.66 [2.79]	7.34 [4.58]
12	-1.74 [-0.97]	0.23 [0.13]	1.04 [0.58]	1.85 [1.04]	2.82 [1.58]	4.63 [2.37]	6.36 [3.58]

Figure 1. Number of available currencies



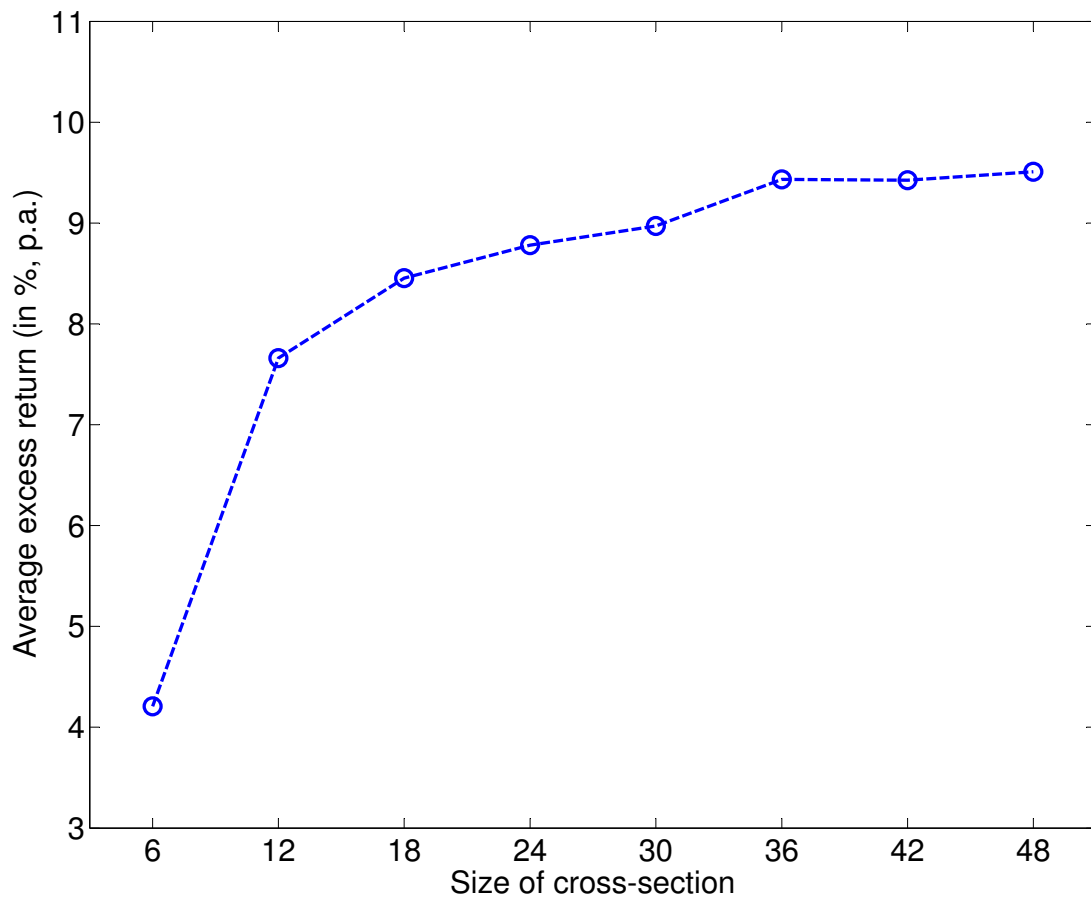
The solid line shows the number of available currencies (i.e., currencies with available data for forward and spot exchange rates) and the dashed line shows the number of currencies with available data when excluding pegged currencies for each month of our sample period from 1976:1 – 2010:1.

Figure 2. Cumulative excess returns of momentum strategies



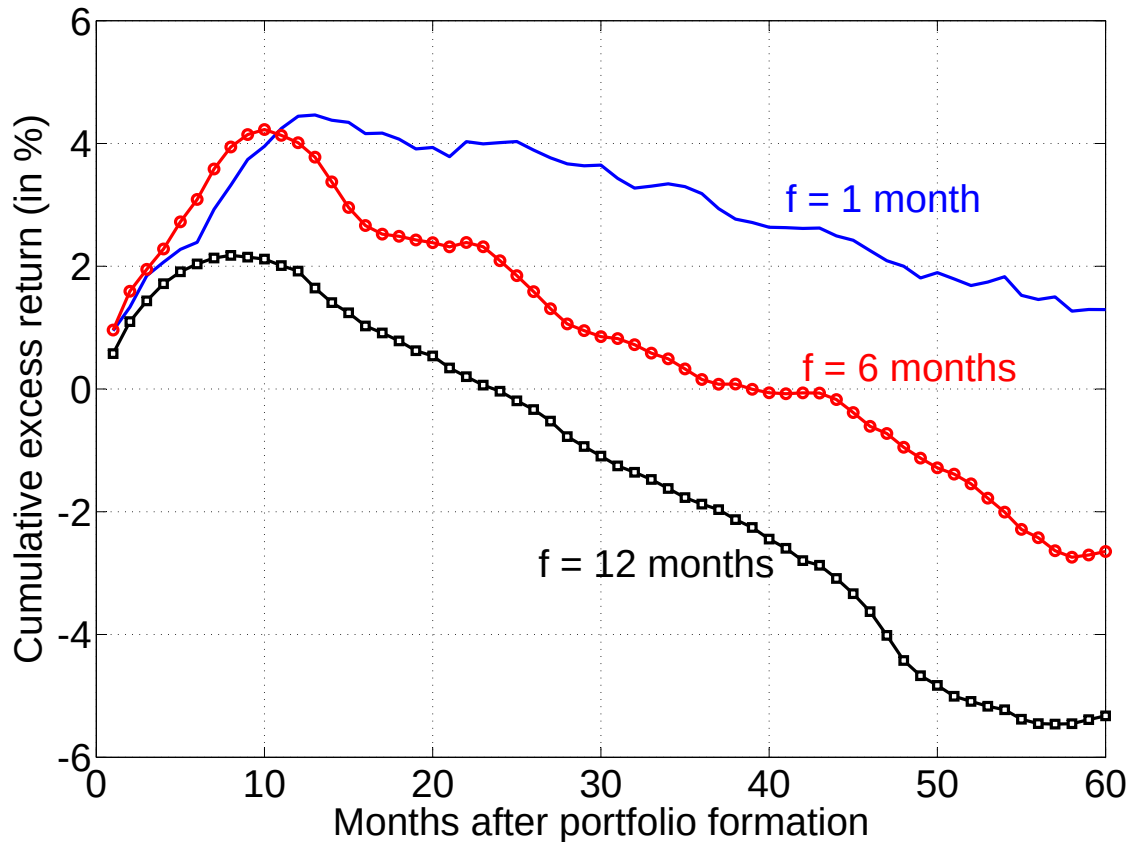
This figure shows cumulative log excess returns (not adjusted for transaction costs) accruing to three different momentum returns. The momentum strategies are for a formation period of 1, 6, and 12 months, respectively, and the holding period is one month. The bold, blue line shows returns to the momentum strategy with a one month formation period (MOM(1,1) in the figure), the dashed, red line shows returns to a strategy with a six months formation period (MOM(6,1)), whereas the thin, black line shows returns to a momentum strategy with a twelve months formation period (MOM(12,1)). Shaded areas correspond to NBER recessions.

Figure 3. Size of the cross-section and momentum returns



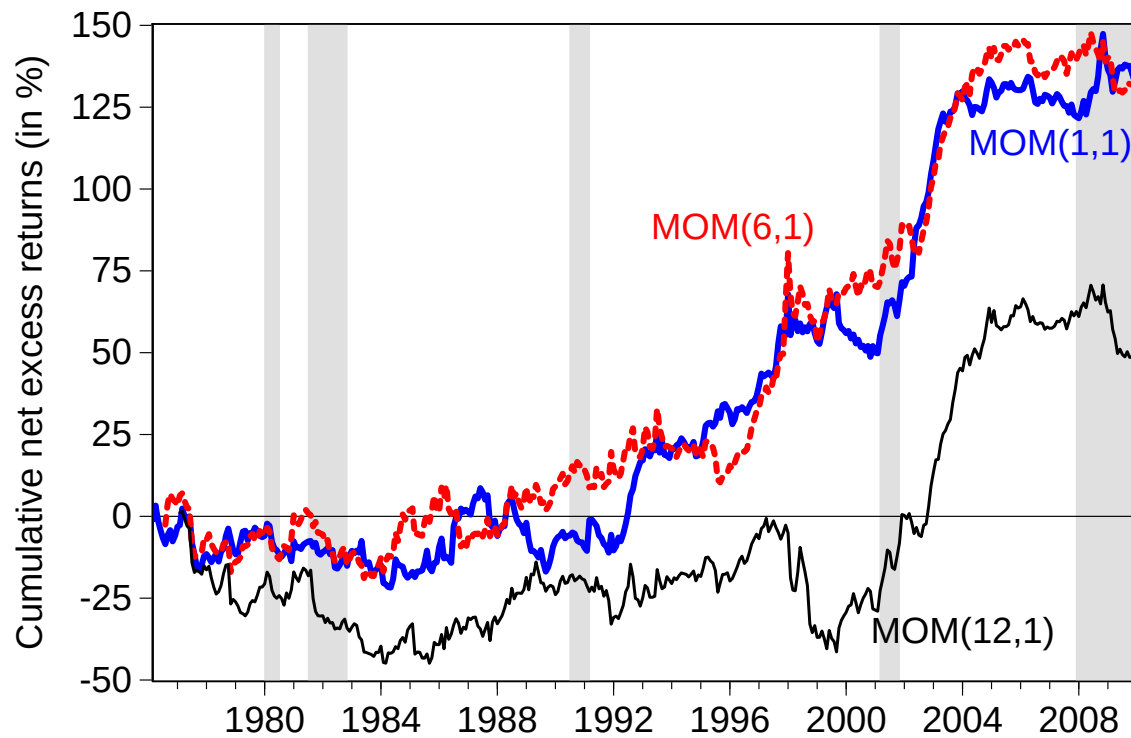
This figure shows average annualized excess returns for a MOM(1,1) strategy implemented on a cross-section of 6, 12, 18, ..., 48 currencies. We draw 5,000 random combinations of currencies for each size of the cross-section (imposing the restriction that we have at least six currencies at each point in time) and plot the average (across simulations) annualized mean excess return of a MOM(1,1) strategy.

Figure 4. Long-horizon momentum excess returns



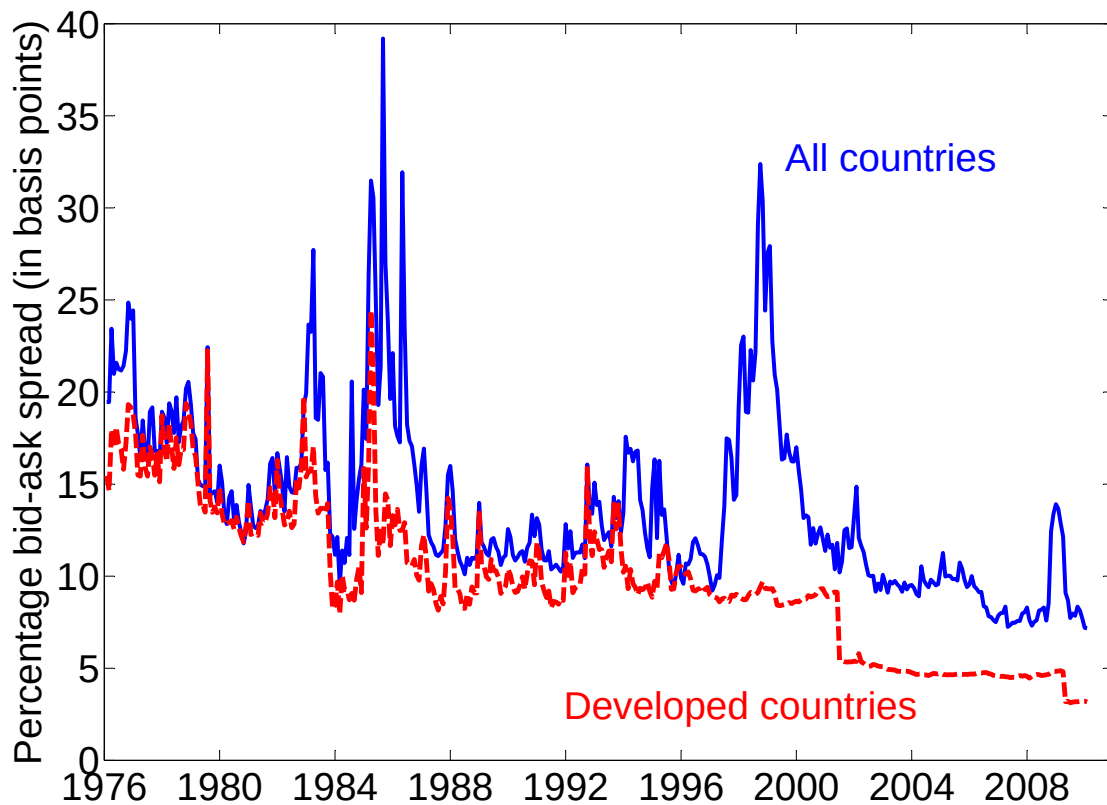
This figure shows cumulative average excess returns to three different long-short currency momentum portfolios after portfolio formation. Momentum portfolios differ in their formation period ($f = 1, 6, 12$ months) and post-formation returns are shown for $1, 2, \dots, 60$ months following the formation period (i.e. we build new portfolios each months but track these portfolios for the first 60 months after their formation so that we are effectively using overlapping horizons). Excess returns are monthly and the sample period is 1976:1 – 2010:1.

Figure 5. Cumulative net excess returns of momentum strategies



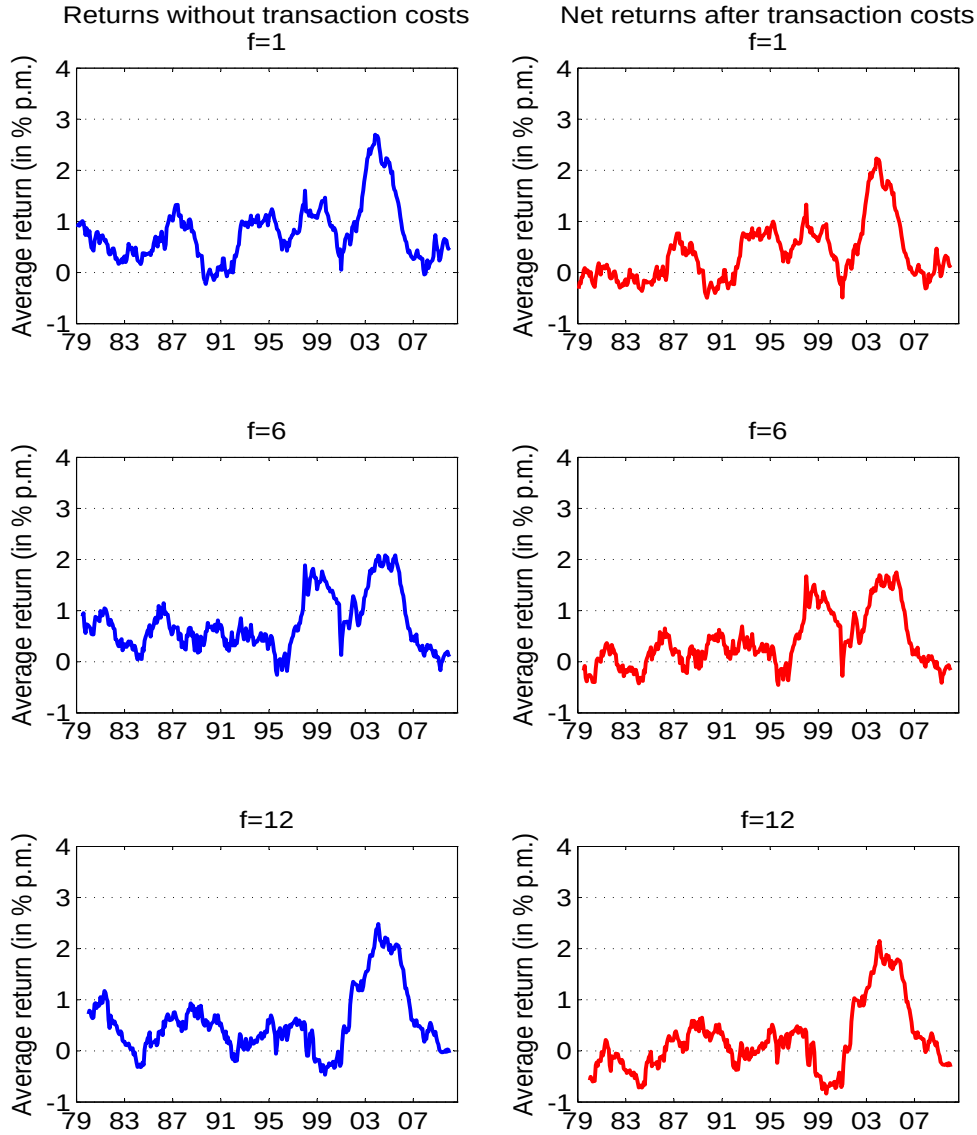
This figure shows cumulative log excess returns adjusted for transaction costs accruing to three different momentum returns. The momentum strategies are for a formation period of 1, 6, and 12 months, respectively, and the holding period is one month. The bold, blue line shows returns to the momentum strategy with a one month formation period (MOM(1,1) in the figure), the dashed, red line shows returns to a strategy with a six months formation period (MOM(6,1)), whereas the thin, black line shows returns to a momentum strategy with a twelve months formation period (MOM(12,1)). Shaded areas correspond to NBER recessions.

Figure 6. Bid-ask spreads over time



This figure shows percentage bid-ask spreads in basis points for the sample period from 1976:1 to 2010:1. The blue solid line shows average spreads for all countries whereas the red dashed line shows spreads for a subset of 15 developed countries. Shown are the average bid-ask spread across countries in a given month and we include both bid-ask spreads between spot rates as well as 1-month forward rates.

Figure 7. Rolling average returns for three momentum strategies



This figure shows average monthly excess returns over rolling windows of 36 months for three long-short momentum strategies: $MOM_{1,1}$, $MOM_{6,1}$ and $MOM_{12,1}$ where $MOM_{j,h}$ denotes the return difference between a portfolio long in currencies with the highest lagged excess returns (measured over the last f months) and a portfolio short in currencies with the lowest excess return over the last f months. Portfolios are held for $h = 1$ month and we use excess returns without transaction costs (left part of the table) and net excess returns adjusted for transaction costs (right part). The sample runs from 1976:1 to 2010:1.

Appendix to accompany
Currency Momentum Strategies

Table A.1. Descriptive statistics: Individual currencies

This table shows descriptive statistics for individual currencies. Means and standard deviations for excess returns and forward discounts are annualized and in percent. Bid-ask spreads are in basis points. The sample period runs from January 1976 to January 2010.

Country	Sample		Excess returns		Forward discounts				Spreads	
	Start	End	mean	std	mean	std	max	min	mean	std
Australia	1984.12	2010.01	0.28	3.35	0.26	0.25	1.37	-0.57	12.92	6.47
Austria	1976.01	1998.12	-0.16	3.33	-0.32	0.28	0.54	-1.70	26.12	19.76
Belgium	1976.01	1998.12	-0.07	3.38	-0.10	0.33	2.23	-1.19	21.67	7.70
Brazil	2004.03	2010.01	1.45	4.49	0.81	0.35	0.69	-0.98	14.43	5.30
Bulgaria	2004.03	2010.01	0.24	3.12	0.06	0.19	1.17	-0.64	6.20	2.08
Canada	1976.01	2010.01	0.02	1.89	0.03	0.21	3.05	-0.35	7.60	2.44
Croatia	2004.03	2010.01	0.42	3.09	0.21	0.31	2.72	-1.12	17.93	6.74
Cyprus	2004.03	2007.12	0.43	2.05	0.01	0.17	1.77	-2.77	21.94	9.54
Czech Rep.	1997.01	2010.01	0.38	3.73	0.13	0.40	1.91	-0.85	13.80	9.41
Denmark	1976.01	2010.01	0.15	3.16	0.12	0.32	0.16	-0.55	13.12	6.92
Egypt	2004.03	2010.01	0.82	0.95	0.64	0.51	0.55	-1.52	43.86	18.60
Euro	1999.01	2010.01	0.12	3.03	-0.03	0.13	1.96	-0.95	4.99	1.83
Finland	1997.01	1998.12	-0.38	2.56	-0.19	0.03	1.93	-0.92	10.44	2.69
France	1976.01	1998.12	0.03	3.15	0.11	0.37	0.54	-1.07	14.31	10.80
Germany	1976.01	1998.12	-0.06	3.31	-0.22	0.34	1.80	-0.89	16.64	14.46
Greece	1997.01	2000.12	-0.31	3.12	0.41	0.25	-0.14	-0.25	11.17	5.69
Hong Kong	1983.1	2010.01	-0.01	0.21	-0.02	0.13	0.59	-2.04	4.10	6.98
Hungary	1997.1	2010.01	0.58	3.92	0.58	0.30	0.86	-0.60	16.74	7.87
Iceland	2004.03	2010.01	-0.20	5.77	0.60	0.20	21.51	-1.88	17.80	18.70
India	1997.1	2010.01	0.11	1.71	0.28	0.20	1.02	-1.80	10.34	8.01
Indonesia	1997.01	2010.01	-0.46	9.56	0.42	3.36	1.21	-1.11	61.67	80.27
Ireland	1976.01	1998.12	-0.31	2.72	0.08	0.24	3.34	-0.28	20.85	21.23
Israel	2004.03	2010.01	0.32	2.69	0.04	0.10	1.19	-0.15	18.67	6.42
Italy	1976.01	1998.12	0.14	3.11	0.42	0.39	1.38	0.07	18.61	11.56
Japan	1978.06	2010.01	-0.09	3.40	-0.30	0.27	1.02	-0.18	12.12	9.68
Kuwait	1997.01	2010.01	0.09	0.58	0.06	0.12	11.19	-17.43	11.45	16.30
Malaysia	1997.01	2010.01	-0.16	4.11	0.04	0.25	0.73	-0.23	6.66	9.84
Mexico	1997.01	2010.01	0.43	2.82	0.75	0.54	1.56	-0.32	9.56	8.47
Netherl.	1976.01	1998.12	-0.04	3.30	-0.16	0.29	2.81	0.14	17.06	16.45
New Z.	1984.12	2010.01	0.52	3.54	0.39	0.40	1.97	-0.03	22.15	15.67
Norway	1976.01	2010.01	0.14	2.94	0.15	0.37	0.72	-0.12	13.41	8.66
Philippines	1997.01	2010.01	0.08	2.85	0.44	0.33	7.62	-0.79	34.78	28.12
Poland	2002.02	2010.01	0.62	4.26	0.22	0.22	0.24	-0.27	14.18	4.60
Portugal	1976.01	1998.12	-0.05	3.36	0.74	1.03	0.34	-0.89	156.33	162.70
Russia	2004.03	2010.01	0.30	2.87	0.39	0.87	3.45	-1.15	6.83	3.23
S. Africa	1983.1	2010.01	0.55	5.19	1.15	2.30	1.00	-0.90	51.61	85.08

Table A.1. *(continued)*

Sample			Excess returns		Forward discounts				Spreads	
Country	Start	End	mean	std	mean	std	max	min	mean	std
S. Korea	2002.02	2010.01	0.18	3.84	0.05	0.20	4.51	-0.20	17.56	18.83
Saudi A.	1997.01	2010.01	0.01	0.13	0.01	0.06	1.46	0.00	2.53	3.69
Singapore	1984.12	2010.01	0.02	1.53	-0.13	0.19	2.02	0.10	18.06	18.92
Slovakia	2002.02	2010.01	0.96	3.46	0.12	0.23	4.42	-0.27	14.48	5.38
Slovenia	2004.03	2006.12	0.25	2.21	0.02	0.15	0.58	-0.19	8.96	2.46
Spain	1976.01	1998.12	0.03	3.30	0.40	0.56	1.31	-0.20	24.61	14.74
Sweden	1976.01	2010.01	0.00	3.16	0.13	0.34	0.43	-0.20	14.85	6.08
Switz.	1976.01	2010.01	-0.07	3.57	-0.29	0.34	0.27	-0.14	16.11	15.88
Taiwan	1997.01	2010.01	-0.17	1.69	-0.07	0.30	1.15	0.10	17.09	11.17
Thai	1997.01	2010.01	0.07	3.76	0.22	0.52	0.35	-0.18	21.43	19.18
UK	1976.01	2010.01	0.12	3.10	0.18	0.24	0.42	-0.22	7.01	4.22
Ukraine	2004.03	2010.01	0.10	4.16	0.69	0.75	4.80	-0.33	61.04	55.17

Table A.2. Momentum returns over subsamples

This table shows average momentum excess returns and Sharpe Ratios for four subsamples of equal length. All five momentum strategies have a holding period of one month ($h = 1$) and the formation period (f) is $f = 1, 3, 6, 9, 12$ months. Numbers in brackets are t -statistics based on Newey and West (1987).

		Formation period f				
		1	3	6	9	12
02/1976 – 07/1984	Mean	6.20	9.58	7.12	4.39	4.67
	t	[1.96]	[2.94]	[2.48]	[1.45]	[1.47]
	SR	0.70	1.02	0.84	0.53	0.53
08/1984 – 01/1993	Mean	7.79	9.86	6.40	5.80	4.96
	t	[2.09]	[3.15]	[2.06]	[1.75]	[1.77]
	SR	0.78	0.84	0.58	0.57	0.50
09/1984 – 07/2001	Mean	10.16	6.59	10.84	7.81	4.72
	t	[3.11]	[1.53]	[2.51]	[1.75]	[1.17]
	SR	0.93	0.53	0.82	0.56	0.41
08/2001 – 01/2010	Mean	12.37	12.83	9.08	11.49	10.10
	t	[3.05]	[3.86]	[2.43]	[3.00]	[2.47]
	SR	1.28	1.42	0.95	1.21	1.04

Table A.3. Momentum for individual currencies

This table reports descriptive statistics (mean, standard deviation, skewness, kurtosis) and Sharpe Ratios of momentum strategies in individual currencies (against the USD) in Panel A. These strategies go long (short) in the foreign currency if last month's return was positive (negative). Panel B reports the average of all individual countries' statistics ("Aver."), the same statistics as in Panel A but for an equally weighted portfolio of all individual strategies ("EW"), and, for comparison, the same statistics for the high minus low portfolio of a cross-sectional momentum strategy (MOM(1,1)).

Panel A. Individual currencies											
	Mean	Std	Skew	Kurt	SR		Mean	Std	Skew	Kurt	SR
Australia	6.18	11.51	0.62	5.04	0.54	Japan	4.80	11.67	0.03	4.49	0.41
Austria	4.53	11.50	-0.07	3.88	0.39	Kuwait	0.73	2.04	2.27	18.88	0.36
Belgium	3.48	11.71	0.14	4.33	0.30	Malaysia	0.69	13.48	-4.83	49.52	0.05
Brazil	7.98	16.29	0.38	3.53	0.49	Mexico	4.63	9.83	0.47	8.23	0.47
Bulgaria	3.76	10.86	-0.69	5.96	0.35	Neth.	5.63	11.34	0.00	4.11	0.50
Canada	0.59	6.55	-0.75	12.62	0.09	New Z.	5.57	12.30	-0.20	4.96	0.45
Croatia	6.05	10.73	0.10	4.76	0.56	Norway	4.65	10.13	0.12	4.47	0.46
Cyprus	-0.57	7.32	0.08	2.66	-0.08	Philipp.	3.57	9.86	-0.36	8.02	0.36
Czech R.	2.19	12.95	-0.35	3.89	0.17	Poland	4.29	14.91	-0.20	5.24	0.29
Denmark	5.95	10.83	-0.12	4.14	0.55	Portugal	2.33	11.66	0.24	6.33	0.20
Egypt	9.44	3.41	1.25	8.69	2.76	Russia	10.91	9.52	2.86	15.05	1.15
Euro	7.58	10.19	-0.45	4.96	0.74	S. Africa	9.76	17.85	0.92	5.00	0.55
Finland	-7.53	8.52	0.13	3.43	-0.88	S. Korea	3.91	13.30	0.21	8.43	0.29
France	2.75	10.89	-0.17	3.88	0.25	Saudi A.	0.12	0.46	-6.06	52.52	0.26
Germany	5.00	11.39	0.01	3.96	0.44	Singapore	1.89	5.28	-0.76	5.76	0.36
Greece	3.20	10.88	-0.48	3.76	0.29	Slovakia	7.17	12.32	-0.54	4.15	0.58
Hong K.	0.37	0.73	0.31	8.79	0.51	Slovenia	2.19	7.76	0.27	2.33	0.28
Hungary	4.86	13.70	-0.50	8.69	0.35	Spain	5.99	11.11	-0.67	6.93	0.54
Iceland	8.50	19.99	1.08	7.18	0.43	Sweden	5.94	10.83	0.56	5.62	0.55
India	5.31	5.50	1.21	6.68	0.97	Switzerl.	3.46	12.33	-0.56	4.73	0.28
Indonesia	25.50	32.42	2.33	17.03	0.79	Taiwan	2.42	5.86	-0.52	6.99	0.41
Ireland	6.08	9.37	-0.14	3.42	0.65	Thailand	3.59	13.02	-1.48	15.83	0.28
Israel	0.90	9.46	-0.37	3.96	0.10	Ukraine	9.35	14.25	2.05	14.58	0.66
Italy	6.67	10.61	-0.17	4.86	0.63	U. Kingdom	4.11	10.68	0.00	4.69	0.39
Panel B. Aggregate statistics											
Aver.	4.86	10.81	-0.06	8.48	0.45						
EW	4.86	6.39	-0.14	5.97	0.76						
MOM(1,1)	9.46	9.92	0.15	5.28	0.95						

Table A.4. Turnover and relative bid-ask spreads of momentum portfolios

This table shows turnover for different momentum portfolios, different combinations of formation (f) and holding (h) periods in Panel A. Numbers are in percent and show the average fraction of portfolio switches (relative to the total number of currencies in a portfolio) per month. We report results for the winner portfolio that contains currencies with the highest lagged excess returns (rows “High”), the loser portfolio that contains the currencies with the lowest lagged returns (rows “Low”), and the average across all six momentum portfolios for a given combination of f and h . Panel B shows relative bid-ask spreads for winner and loser portfolios. We report average bid-ask spreads (in basis points) in excess of the cross-sectional average bid-ask spread of all currencies in a given month. The sample period runs from January 1976 to January 2010.

Panel A: Turnover							Panel B: Bid-ask spreads						
f	PF	Holding period h					f	PF	Holding period h				
		1	3	6	9	12			1	3	6	9	12
1	High	74.3	24.5	12.2	7.9	5.9	1	High	2.6	1.4	1	0.1	0.8
	Low	72.2	26.0	13.1	8.8	6.5		Low	3.1	2.1	1.4	0.4	0.8
	All	77.8	26.3	13.4	8.6	6.4							
3	High	42.4	24.2	12.8	7.9	6.1	3	High	2.7	0.3	0.8	0	0.9
	Low	43.8	24.9	12.9	8.8	6.3		Low	2.6	0.6	-0.1	-0.2	0.1
	All	59.1	26.3	13.0	8.8	6.5							
6	High	29.9	17.7	12.6	8.4	6.8	6	High	2.6	1	0.4	0.9	0.1
	Low	31.1	17.6	12.3	8.4	6.7		Low	4.1	0.6	0.1	0.4	0.4
	All	48.4	22.3	13.0	8.6	6.7							
9	High	23.6	13.8	9.9	8.3	6.5	9	High	3.5	1.4	1.2	0.2	0.5
	Low	24.0	14.3	9.8	7.7	6.4		Low	5.3	1.6	0.6	0.7	0.4
	All	40.3	19.1	11.7	8.5	6.6							
12	High	21.9	12.0	9.2	8.1	6.5	12	High	3.3	0.9	0.3	0	0.1
	Low	20.3	12.5	8.8	6.8	6.0		Low	6.9	2.4	0.9	0.6	0.6
	All	37.2	18.0	11.4	8.4	6.6							

Table A.5. Portfolio belongings

This table reports the share of months in which a country is included in the portfolio with lowest lagged returns (Portfolio Low) and the portfolio with highest lagged returns (Portfolio High). Results are based on the strategy with a formation and holding period of one month (MOM(1,1)).

Country	Low	High	Country	Low	High
Australia	0.18	0.19	Japan	0.32	0.21
Austria	0.08	0.02	Kuwait	0.04	0.02
Belgium	0.07	0.05	Malaysia	0.06	0.04
Brazil	0.03	0.09	Mexico	0.07	0.09
Bulgaria	0.01	0.01	Netherlands	0.07	0.05
Canada	0.24	0.26	New Zealand	0.15	0.21
Croatia	0.02	0.02	Norway	0.11	0.11
Cyprus	0.00	0.00	Philippines	0.07	0.08
Czech Republic	0.08	0.10	Poland	0.06	0.08
Denmark	0.08	0.11	Portugal	0.05	0.08
Egypt	0.00	0.03	Russia	0.01	0.01
Euro	0.03	0.02	South Africa	0.20	0.21
Finland	0.00	0.00	South Korea	0.04	0.04
France	0.05	0.05	Saudi A.	0.05	0.04
Germany	0.08	0.06	Singapore	0.11	0.05
Greece	0.01	0.02	Slovakia	0.03	0.05
Hong Kong	0.18	0.11	Slovenia	0.00	0.00
Hungary	0.04	0.08	Spain	0.07	0.12
Iceland	0.05	0.05	Sweden	0.12	0.13
India	0.04	0.05	Switzerland	0.27	0.18
Indonesia	0.12	0.10	Taiwan	0.07	0.02
Ireland	0.05	0.06	Thailand	0.05	0.06
Israel	0.02	0.03	United Kingdom	0.16	0.20
Italy	0.05	0.12	Ukraine	0.02	0.04

Table A.6. Momentum returns: Different base currencies

This table shows annualized average excess returns for different momentum strategies $(\overline{r^{f,h}})$ as in Table 1 but here we compute excess returns from the perspective of a non-U.S. investor, i.e., we change the base currency from U.S. dollars to British Pound (GBP), Swiss Franc (CHF), Canadian dollar (CAD), or Swedish kronor (SEK).

Holding period h						Holding period h					
	1	3	6	9	12	f	1	3	6	9	12
f	Excess returns (GBP)						Excess returns (CHF)				
1	9.44	6.89	6.17	5.15	5.75	1	9.40	6.86	6.04	5.09	5.31
	[5.32]	[4.05]	[3.13]	[2.73]	[3.16]		[5.37]	[4.07]	[3.08]	[2.72]	[2.94]
3	9.38	6.36	5.05	4.92	4.66	3	9.47	6.39	4.93	4.76	4.41
	[5.28]	[3.80]	[3.08]	[3.06]	[2.87]		[5.34]	[3.85]	[3.03]	[2.98]	[2.74]
6	8.58	6.56	3.74	3.66	3.34	6	8.49	6.21	3.69	3.26	3.14
	[4.79]	[3.79]	[2.10]	[2.02]	[1.82]		[4.74]	[3.58]	[2.09]	[1.79]	[1.70]
9	7.28	6.93	5.45	3.89	3.50	9	7.09	6.72	5.43	3.76	3.47
	[3.86]	[3.71]	[2.91]	[2.07]	[1.79]		[3.78]	[3.63]	[2.90]	[2.01]	[1.77]
12	6.20	5.52	3.07	2.00	1.80	12	6.22	5.41	2.99	2.05	1.98
	[3.43]	[3.26]	[1.79]	[1.14]	[1.00]		[3.45]	[3.21]	[1.75]	[1.18]	[1.10]
Excess returns (CAD)						Excess returns (SEK)					
f	1	3	6	9	12	f	1	3	6	9	12
1	9.27	6.81	5.96	5.01	5.32	1	9.48	6.98	6.12	5.09	5.63
	[5.23]	[4.06]	[3.05]	[2.68]	[2.94]		[5.33]	[4.06]	[3.09]	[2.72]	[3.09]
3	9.49	6.43	4.98	4.67	4.43	3	9.44	6.37	4.98	4.79	4.43
	[5.33]	[3.86]	[3.04]	[2.92]	[2.74]		[5.32]	[3.83]	[3.04]	[2.99]	[2.74]
6	8.57	6.43	3.61	3.27	3.15	6	8.58	6.30	3.76	3.27	3.20
	[4.77]	[3.68]	[2.02]	[1.80]	[1.70]		[4.76]	[3.63]	[2.11]	[1.80]	[1.73]
9	7.16	6.81	5.54	3.90	3.37	9	7.18	6.74	5.46	3.90	3.21
	[3.79]	[3.66]	[2.93]	[2.06]	[1.73]		[3.81]	[3.61]	[2.90]	[2.06]	[1.65]
12	6.18	5.44	3.04	2.06	1.96	12	6.17	5.46	3.14	2.04	1.96
	[3.40]	[3.20]	[1.76]	[1.17]	[1.08]		[3.41]	[3.22]	[1.82]	[1.17]	[1.08]

Table A.7. Momentum returns: Exchange rate changes for different base currencies

This table shows annualized average spot exchange rate changes for different momentum strategies ($\overline{r^{f,h}}$) as in Table 1 but here we compute spot rate changes from the perspective of a non-U.S. investor, i.e., we change the base currency from U.S. dollars to British Pound (GBP), Swiss Franc (CHF), Canadian dollar (CAD), or Swedish kronor (SEK).

Holding period h						Holding period h					
	1	3	6	9	12	f	1	3	6	9	12
f	Spot rate changes (GBP)						Spot rate changes (CHF)				
1	7.95	4.74	3.67	4.47	2.98	1	7.86	5.01	4.33	4.37	4.11
	[4.59]	[3.26]	[2.10]	[2.62]	[1.79]		[4.56]	[3.52]	[2.49]	[2.68]	[2.61]
3	9.31	6.27	5.49	5.36	5.66	3	8.96	5.58	5.24	4.88	5.81
	[5.35]	[3.92]	[3.74]	[3.16]	[3.72]		[5.33]	[3.53]	[3.66]	[2.97]	[3.88]
6	7.06	6.18	4.36	4.54	3.68	6	7.03	5.92	4.08	4.37	3.53
	[4.11]	[4.15]	[2.57]	[2.69]	[2.02]		[4.22]	[4.03]	[2.53]	[2.63]	[2.03]
9	8.90	8.07	7.42	5.52	5.40	9	8.76	7.67	6.65	5.27	4.51
	[5.05]	[4.74]	[4.40]	[3.24]	[2.94]		[5.14]	[4.61]	[4.05]	[3.18]	[2.59]
12	8.05	6.69	4.65	3.10	3.30	12	7.64	6.10	4.11	3.65	3.06
	[4.68]	[4.25]	[2.91]	[1.82]	[1.96]		[4.60]	[4.02]	[2.72]	[2.26]	[1.94]
Spot rate changes (CAD)							Spot rate changes (SEK)				
f	1	3	6	9	12	f	1	3	6	9	12
1	8.57	5.02	3.56	5.06	3.24	1	7.84	4.97	3.72	5.42	3.48
	[4.98]	[3.41]	[2.01]	[3.02]	[2.08]		[4.51]	[3.47]	[2.11]	[3.28]	[2.23]
3	8.65	5.55	5.45	4.71	5.32	3	9.46	5.80	5.31	4.97	5.61
	[5.10]	[3.52]	[3.81]	[2.99]	[3.67]		[5.46]	[3.60]	[3.68]	[3.02]	[3.77]
6	7.16	5.47	3.53	4.17	3.26	6	6.68	5.65	3.78	4.48	3.68
	[4.23]	[3.76]	[2.13]	[2.63]	[1.80]		[3.95]	[3.83]	[2.34]	[2.74]	[2.09]
9	8.76	7.47	6.57	5.15	4.46	9	8.64	7.61	6.68	4.89	4.55
	[5.07]	[4.51]	[4.00]	[3.01]	[2.52]		[4.95]	[4.51]	[4.01]	[2.92]	[2.55]
12	7.58	5.92	3.88	3.40	3.02	12	8.04	6.39	4.68	3.61	3.54
	[4.54]	[3.87]	[2.56]	[2.14]	[1.90]		[4.73]	[4.08]	[3.02]	[2.21]	[2.17]

Table A.8. Macro risk for other base currencies

This table shows regressions of momentum returns (MOM(1,1)-strategy) for different base currencies (see Table A.6) on macro risk factors of the respective country. These risk factors are growth in industrial production (IP), CPI inflation (INF), growth in real money balances (narrow money, M), changes in terms spreads (10 year minus 3 month maturity, TS), and local stock market returns (Datastream country stock market indices, S). We report the intercept (α), slope coefficient (β), t-statistics (in brackets), and the R^2 (in percent) of univariate regressions of returns on one of these risk factors. The sample period is 1976 to 2010 and the frequency is monthly.

	IP	INF	M	TS	S
United Kingdom					
α	0.77	0.73	0.77	0.83	0.77
	[5.01]	[4.61]	[5.02]	[4.35]	[5.02]
β	1.71	32.99	0.71	0.25	-2.45
	[0.17]	[1.15]	[0.23]	[0.24]	[-0.75]
$R^2(\%)$	0.01	0.31	0.01	0.04	0.20
Switzerland					
α	1.14	0.87	0.86	0.76	0.83
	[4.47]	[4.87]	[4.75]	[5.01]	[3.97]
β	-73.85	-56.97	7.54	-0.26	-1.64
	[-2.58]	[-1.49]	[0.35]	[-0.87]	[-0.30]
$R^2(\%)$	3.59	0.53	0.04	0.16	0.08
Canada					
α	1.01	0.78	0.75	0.76	0.76
	[3.77]	[4.09]	[4.92]	[5.01]	[4.66]
β	-20.93	-3.41	13.46	0.03	1.51
	[-0.71]	[-0.10]	[1.03]	[0.23]	[0.36]
$R^2(\%)$	0.42	0.00	0.30	0.01	0.07
Sweden					
α	0.78	0.74	0.77	0.77	0.82
	[4.89]	[4.07]	[5.04]	[5.03]	[4.68]
β	-1.93	9.83	-1.89	0.02	-3.60
	[-0.38]	[0.51]	[-0.45]	[0.09]	[-1.08]
$R^2(\%)$	0.03	0.04	0.05	0.00	0.48

Table A.9. Momentum and currency regimes

This table shows average excess returns for momentum portfolios when we restrict our universe of currencies to managed floats and floating currencies (Panel A) or only floating currencies (Panel B). We report average excess returns for six portfolios sorted on lagged one, six, and twelve month returns, and the high minus low momentum portfolio (H-L). Numbers in brackets are t -statistics based on Newey and West (1987). The sample starts in 1986 to obtain a sufficiently broad cross-section of floating currencies.

Panel A. Managed floats and floating currencies							
f	L	2	3	4	5	H	H-L
1	-4.17 [-1.59]	1.18 [0.57]	2.28 [1.29]	2.43 [1.07]	2.93 [1.49]	9.02 [3.48]	13.19 [4.40]
6	-2.16 [-0.68]	1.10 [0.58]	1.54 [0.87]	2.77 [1.46]	2.65 [1.28]	8.09 [2.97]	10.25 [3.07]
12	0.33 [0.11]	-0.09 [-0.05]	1.32 [0.70]	2.69 [1.45]	5.45 [2.67]	6.98 [2.83]	6.65 [2.02]
Panel B. Floating currencies							
f	L	2	3	4	5	H	H-L
1	-1.28 [-0.51]	-0.14 [-0.07]	1.50 [0.72]	0.82 [0.36]	1.43 [0.65]	8.22 [2.81]	9.50 [2.76]
6	0.12 [0.05]	-0.71 [-0.38]	0.44 [0.22]	1.56 [0.86]	2.30 [1.08]	7.60 [2.35]	7.48 [2.27]
12	-0.51 [-0.19]	-1.18 [-0.61]	1.28 [0.64]	1.53 [0.83]	3.51 [1.58]	8.21 [2.82]	8.72 [2.47]

Table A.10. Momentum returns and central bank interventions

This table reports results for regressions of momentum returns (MOM(1,1), MOM(6,1), MOM(12,1)) on central bank intervention activity. Central bank interventions are calculated as the sum of absolute intervention volumes (in 100 million dollars) within each month and we consider all interventions in the USD based on data from the FRED database (Fed St. Louis). We include contemporaneous intervention volumes (cb), and two lags of intervention volumes. The sample period is 1976 to 2010 and the frequency is monthly.

	MOM(1,1)	MOM(6,1)	MOM(12,1)
const.	0.93 [5.02]	0.78 [4.18]	0.50 [2.68]
cb	-0.03 [-2.02]	0.01 [0.67]	0.03 [1.37]
cb_{t-1}	-0.01 [-0.86]	-0.02 [-1.08]	-0.02 [-0.99]
cb_{t-2}	-0.01 [-0.76]	-0.01 [-1.01]	0.00 [-0.24]
$\overline{R}^2(\%)$	0.43	-0.28	-0.18

Table A.11. Momentum returns since 1992

This table is identical to Table 1 but here the sample period is January 1992 – January 2010 so that we are looking at a period where bid-ask spreads are significantly lower than in the very early part of our sample.

Excess returns (without b/a)						Spot rate changes (without b/a)					
f	Holding period h					f	Holding period h				
	1	3	6	9	12		1	3	6	9	12
1	11.69	7.74	8.15	5.08	7.45	1	7.88	2.80	1.82	2.18	3.99
	[4.54]	[3.06]	[2.73]	[1.85]	[2.58]		[3.27]	[1.41]	[0.72]	[1.01]	[1.57]
3	9.95	8.12	7.82	3.25	6.79	3	6.90	5.99	5.65	2.64	5.63
	[3.64]	[3.11]	[3.09]	[1.29]	[2.66]		[2.78]	[2.64]	[2.63]	[1.11]	[2.42]
6	9.96	7.99	5.74	5.26	4.41	6	6.02	4.77	2.33	3.63	3.61
	[3.51]	[2.82]	[2.13]	[1.84]	[1.55]		[2.53]	[2.52]	[1.07]	[1.61]	[1.54]
9	9.77	8.59	7.08	5.20	1.93	9	8.47	6.36	6.12	4.68	2.62
	[3.25]	[2.87]	[2.34]	[1.66]	[0.72]		[3.54]	[2.65]	[2.46]	[1.90]	[1.26]
12	7.04	7.18	4.12	2.95	1.70	12	6.66	5.66	2.64	1.21	0.34
	[2.36]	[2.60]	[1.47]	[1.02]	[0.61]		[2.66]	[2.61]	[1.30]	[0.53]	[0.17]
Excess returns (with b/a)						Spot rate changes (with b/a)					
f	Holding period h					f	Holding period h				
	1	3	6	9	12		1	3	6	9	12
1	7.27	4.21	4.87	2.07	4.33	1	4.55	1.49	1.06	1.81	3.70
	[2.85]	[1.70]	[1.68]	[0.75]	[1.54]		[1.89]	[0.77]	[0.43]	[0.84]	[1.46]
3	6.03	4.72	4.68	0.33	3.82	3	5.08	4.95	5.19	2.22	5.32
	[2.21]	[1.77]	[1.85]	[0.13]	[1.48]		[2.00]	[2.14]	[2.39]	[0.92]	[2.28]
6	6.41	4.70	2.86	2.16	1.51	6	4.69	4.05	1.74	3.30	3.41
	[2.28]	[1.64]	[1.05]	[0.75]	[0.52]		[1.96]	[2.13]	[0.80]	[1.47]	[1.45]
9	6.35	5.29	3.87	1.90	-0.86	9	7.53	5.80	5.73	4.39	2.47
	[2.07]	[1.76]	[1.27]	[0.59]	[-0.31]		[3.14]	[2.40]	[2.30]	[1.78]	[1.18]
12	3.79	4.00	1.12	0.08	-0.85	12	5.80	5.16	2.40	1.00	0.27
	[1.25]	[1.43]	[0.39]	[0.03]	[-0.30]		[2.31]	[2.36]	[1.17]	[0.44]	[0.13]

Table A.12. Momentum returns in developed countries

This setup of this table is identical to Table 1 but here we show results for a smaller sub-sample of 15 developed countries as defined in the main text.

Excess returns						Spot rate changes					
f	Holding period h					f	Holding period h				
	1	3	6	9	12		1	3	6	9	12
1	3.83	4.79	3.88	2.95	2.24	1	2.89	3.67	4.44	3.05	2.82
	[2.72]	[3.27]	[2.00]	[1.40]	[1.39]		[1.60]	[2.28]	[2.54]	[1.64]	[1.70]
3	5.71	3.85	2.26	2.60	2.42	3	4.94	2.74	1.83	2.03	1.21
	[3.58]	[1.68]	[1.97]	[0.76]	[2.17]		[2.99]	[1.63]	[1.07]	[1.30]	[0.77]
6	3.70	2.59	1.83	1.85	-0.14	6	2.14	2.50	1.91	1.87	0.27
	[2.46]	[1.49]	[1.91]	[1.12]	[0.63]		[1.23]	[1.47]	[1.18]	[1.02]	[0.15]
9	3.96	3.35	2.04	1.36	-0.82	9	4.04	4.06	3.42	3.10	0.98
	[1.61]	[1.63]	[1.32]	[0.85]	[-0.65]		[2.14]	[2.22]	[1.92]	[1.77]	[0.53]
12	3.14	2.98	0.54	1.27	-0.16	12	3.06	2.69	1.28	1.63	0.55
	[1.84]	[2.02]	[1.14]	[1.66]	[0.77]		[1.63]	[1.53]	[0.75]	[1.03]	[0.33]

Table A.13. Momentum returns in developed countries after transaction costs

This setup of this table is identical to Table 1 but here we show results for a smaller sub-sample of 15 developed countries as defined in the main text.

Net excess returns						Net spot rate changes					
f	Holding period h					f	Holding period h				
	1	3	6	9	12		1	3	6	9	12
1	0.79	2.11	1.38	1.50	0.50	1	0.86	3.00	4.14	2.83	2.58
	[0.44]	[1.23]	[0.77]	[0.79]	[0.29]		[0.47]	[1.83]	[2.39]	[1.52]	[1.55]
3	3.32	1.02	-1.23	1.31	-0.49	3	3.69	2.05	1.46	1.82	1.04
	[2.05]	[0.61]	[-0.73]	[0.79]	[-0.30]		[2.23]	[1.21]	[0.86]	[1.14]	[0.66]
6	1.96	0.83	-0.47	0.15	-1.87	6	1.33	2.03	1.59	1.65	0.11
	[1.18]	[0.49]	[-0.27]	[0.08]	[-1.04]		[0.76]	[1.19]	[0.94]	[0.87]	[0.06]
9	1.59	1.30	0.26	0.08	-3.55	9	3.38	3.66	3.15	2.88	0.79
	[0.89]	[0.76]	[0.14]	[0.05]	[-1.89]		[1.78]	[2.00]	[1.75]	[1.60]	[0.42]
12	1.25	1.05	-1.68	-0.17	-1.62	12	2.42	2.39	1.14	1.45	0.73
	[0.70]	[0.62]	[-0.95]	[-0.11]	[-1.01]		[1.29]	[1.36]	[0.65]	[0.89]	[0.42]

Table A.14. Momentum returns in developed countries starting in 1992

This setup of this table is identical to Table A.12 but here we show results for developed countries of the sample period 1992 – 2010.

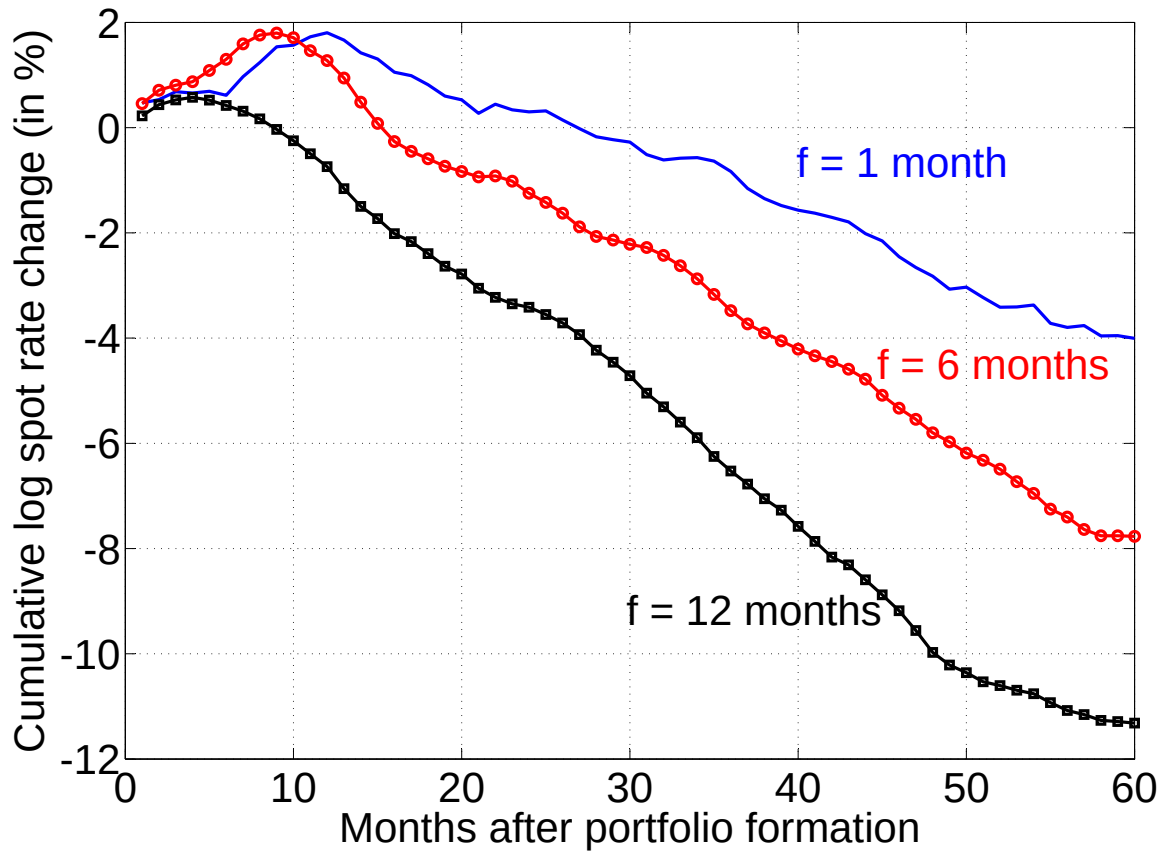
Excess returns (without b/a)						Spot rate changes (without b/a)					
f	Holding period h					f	Holding period h				
	1	3	6	9	12		1	3	6	9	12
1	1.56	4.34	2.76	0.55	2.94	1	0.08	2.76	1.80	-0.28	2.37
	[1.04]	[2.07]	[0.51]	[-0.12]	[1.11]		[0.03]	[1.22]	[0.79]	[-0.12]	[0.91]
3	3.62	2.39	1.70	-0.21	3.38	3	2.35	1.13	0.83	0.29	4.27
	[2.01]	[0.61]	[1.67]	[-0.88]	[1.18]		[1.06]	[0.52]	[0.38]	[0.15]	[2.07]
6	0.80	0.31	2.26	3.30	4.95	6	-0.96	-0.25	1.54	2.66	4.47
	[0.86]	[0.11]	[1.18]	[0.89]	[1.38]		[-0.42]	[-0.11]	[0.70]	[1.02]	[1.70]
9	3.13	1.84	1.22	0.89	1.35	9	1.48	1.27	1.99	1.63	1.93
	[0.82]	[0.65]	[0.83]	[0.23]	[0.52]		[0.58]	[0.52]	[0.77]	[0.70]	[0.90]
12	2.27	2.41	0.97	1.87	0.35	12	1.20	1.89	0.89	0.81	0.94
	[0.89]	[1.27]	[1.07]	[1.87]	[0.48]		[0.47]	[0.77]	[0.38]	[0.38]	[0.41]
Excess returns (with b/a)						Spot rate changes (with b/a)					
f	Holding period h					f	Holding period h				
	1	3	6	9	12		1	3	6	9	12
1	-0.73	2.16	0.79	-0.27	1.38	1	-1.50	2.26	1.65	-0.48	2.25
	[-0.32]	[0.93]	[0.34]	[-0.10]	[0.54]		[-0.65]	[0.96]	[0.74]	[-0.19]	[0.85]
3	1.38	-0.24	-1.82	-0.84	2.20	3	1.32	0.60	0.57	0.13	4.15
	[0.61]	[-0.10]	[-0.73]	[-0.35]	[0.90]		[0.59]	[0.28]	[0.26]	[0.06]	[1.87]
6	-1.19	-0.45	0.84	2.29	4.70	6	-1.55	-0.55	1.35	2.50	4.34
	[-0.53]	[-0.20]	[0.35]	[0.80]	[1.58]		[-0.68]	[-0.25]	[0.56]	[0.89]	[1.50]
9	1.29	0.45	0.35	0.51	0.46	9	1.03	0.96	1.82	1.48	1.64
	[0.52]	[0.19]	[0.13]	[0.20]	[0.18]		[0.40]	[0.40]	[0.69]	[0.59]	[0.71]
12	1.23	1.30	-1.06	1.37	-1.14	12	0.72	1.77	0.93	0.66	0.81
	[0.48]	[0.53]	[-0.42]	[0.63]	[-0.51]		[0.28]	[0.72]	[0.37]	[0.29]	[0.33]

Table A.15. Comparing momentum and carry trade portfolios: Risk characteristics

This Table shows portfolio excess returns for a momentum strategy with a one month formation and holding period (Panel A) as well as for the carry trade strategy (Panel B). For each portfolio of the two strategies, we report the average value of the country risk rating (*CRISK*) and exchange rate stability risk rating (*XSTAB*) at the time of portfolio formation. The risk ratings for each country are relative to the risk rating of the U.S. (deviation in %) and a higher value indicates higher risk.

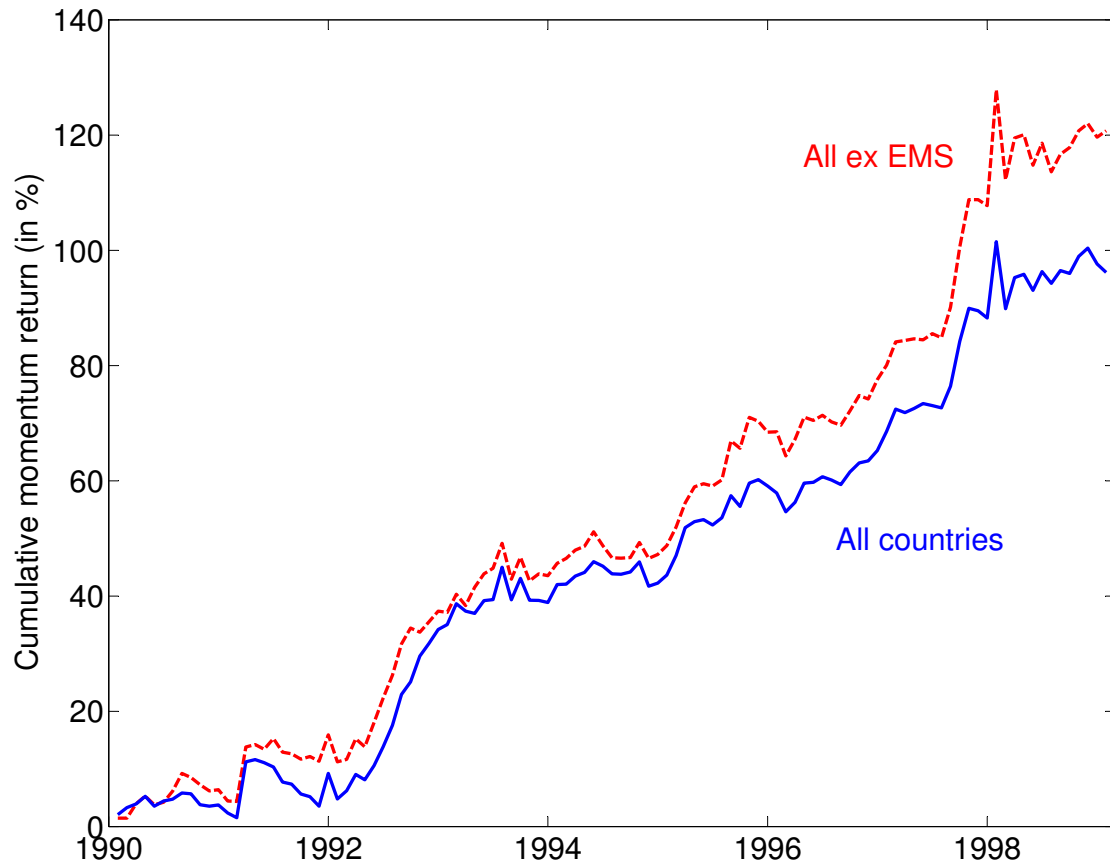
Panel A: Momentum Portfolios ($f = 1, h = 1$)							
	Low	2	3	4	5	High	H-L
<i>CRISK</i>	2.71 [3.91]	0.96 [1.91]	0.52 [1.07]	0.71 [1.37]	1.25 [2.65]	3.58 [5.76]	0.87 [1.44]
<i>XSTAB</i>	3.19 [0.56]	-0.25 [-0.05]	-1.03 [-0.18]	-0.57 [-0.10]	-0.13 [-0.02]	2.72 [0.47]	-0.47 [-0.47]
Panel B: Carry Trade Portfolios							
	Low	2	3	4	5	High	H-L
<i>CRISK</i>	-7.15 [-12.99]	-5.36 [-8.57]	-2.53 [-3.90]	-1.99 [-3.22]	0.44 [0.68]	4.72 [8.44]	11.87 [20.47]
<i>XSTAB</i>	-7.66 [-1.30]	-3.97 [-0.74]	-1.67 [-0.30]	-0.48 [-0.09]	1.59 [0.27]	5.47 [0.91]	13.13 [12.47]

Figure A.1. Long-horizon spot rate changes of momentum portfolios



This figure shows cumulative average spot rate changes to three different long-short currency momentum portfolios after portfolio formation. Momentum portfolios differ in their formation period ($f = 1, 6, 12$ months) and post-formation returns are shown for $1, 2, \dots, 60$ months following the formation period (i.e. we build new portfolios each months but track these portfolios for the first 60 months after their formation so that we are effectively using overlapping horizons). Spot rate changes are monthly and the sample period is 1976:1 – 2010:1.

Figure A.2. Excluding EMS member countries



This figure shows cumulative momentum excess returns ($MOM(1,1)$) over 1990s for the full set of countries (blue, solid line) and for a subset countries that excludes all EMS member countries except Germany.